

What do Interest Rates Reveal about the Stock Market?

A Noisy Rational Expectations Model of Stock and Bond Markets^{*}

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Abstract

We provide novel insights into how investors use information contained in interest rates to learn about economic fundamentals and how this affects informational and allocative efficiency. Specifically, we develop a noisy rational expectations equilibrium model with an endogenous interest rate that investors use to update their beliefs. The model yields two key findings. First, the interest rate reveals primarily information about discount rates, allowing investors to extract more information about cashflows from stock prices. Second, the precision of the interest-rate signal and, hence, stock-price informativeness are positively correlated with the interest rate. We present evidence consistent with this prediction.

Keywords: (endogenous) interest rates, information choice, informational efficiency, rational expectations

JEL: E43, E44, G11, G14

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I. Introduction

Interest rates play an essential role in financial markets. Foremost, they determine the rates at which investors discount future cashflows. But they also convey valuable information about the economic outlook. Recently, however, many market participants have expressed concerns that low interest rates (due to, e.g., an unconventional monetary policy, a global savings glut, an ageing population, or rising income inequality) have distorted the prices of various assets—to the point that their prices have lost their predictive power and capital is misallocated.¹

The purpose of this paper is to provide novel theoretical and empirical insights into the link between long-term interest rates and informational efficiency—the ability of financial markets to aggregate and disseminate private information—as well as real efficiency—their ability to allocate capital. We start by briefly examining the data, with a focus on the U.S. stock market. Indeed, we find that stock-price informativeness (measured as in [Bai, Philippon, and Savov 2016](#)) positively correlates with long-term interest rates, as illustrated in Figure 1 below. Moreover, consistent with this relation, price informativeness tends to increase in the supply of Treasury bonds and to decrease in the demand for Treasury bonds, a finding that lends initial empirical support to claims that low interest rates might reduce the discriminatory power of asset prices.

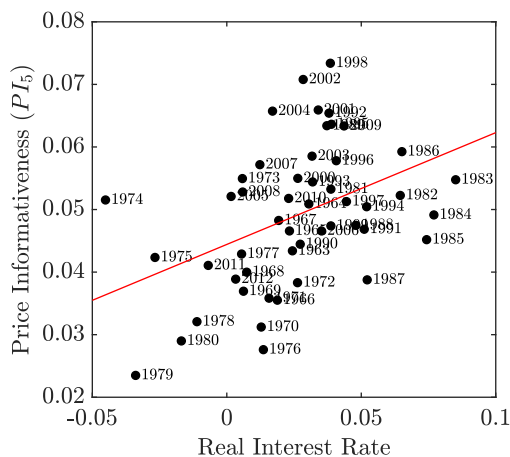


FIGURE 1. STOCK-PRICE INFORMATIVENESS AND THE REAL INTEREST RATE

Notes: The figure plots stock-price informativeness against the long-term real interest rate. Stock-price informativeness is measured as in [Bai, Philippon, and Savov \(2016\)](#) and captures the extent to which firms' current stock prices reflect their future (five-year-ahead) cashflows. The data come from the United States and span the period from 1962 to 2017.

¹For instance, both Jerome Powell, the chairman of the Federal Reserve, and Mario Draghi, the former chairman of the ECB, have raised such concerns ([Draghi 2015](#), [Powell 2017](#)).

The remainder of the paper is dedicated to understanding the theoretical underpinnings of these empirical patterns. For that purpose, we develop a novel noisy rational expectations equilibrium (REE) model of the stock market. The model differs from traditional REE models, such as those of [Grossman and Stiglitz \(1980\)](#) and [Hellwig \(1980\)](#), along one key dimension: we relax the traditional assumption that the bond is in perfectly elastic supply, with an exogenously-fixed interest rate (which rules out any learning from the interest rate). Instead, we assume that the bond supply is imperfectly elastic (e.g., fixed). As a consequence, the equilibrium interest rate now plays a *dual role*: it determines the discount rate and reveals information to investors.

Formally, this change manifests in the following three features that distinguish our model from other REE models of the stock market. First, the interest rate is endogenously determined. Second, investors learn not only from their private signals and the stock price but also from the interest rate. Third, investors make intertemporal consumption choices. Otherwise, the model is kept parsimonious to illustrate the economic mechanisms in the clearest possible way. That is, we consider a two-period model with a continuum of risk-averse investors who receive private signals about the fundamental. They trade a risk-free bond and a risky stock in competitive markets. Noise traders, operating independently in both the stock and the bond market, prevent asset prices from being perfectly revealing. Finally, to illustrate the implications for allocative efficiency, we endogenize output and explicitly model the investment decision of the firm underlying the stock.

Our model yields two set of results. First, it sheds light on *how* and *what type of information* investors learn from the bond market. In brief, the interest rate (or, more precisely, the bond-market signal) reveals information primarily about noise traders' demand in the *stock* market, that is, it provides *discount-rate news*.² Combined with the stock's price, this information allows investors to form more precise beliefs about cashflows.

This result is a consequence of budget constraints and market clearing. To see this, assume first that investors only consume at the terminal date.³ Investors' budget constraints then imply that their aggregate bond and stock expenditures add up to their aggregate wealth. In addition, market clearing in the stock market requires that investors' aggregate stock demand equals the stock supply minus noise traders' demand.

²In line with the literature, we interpret noise-trader shocks as discount rate news because these shocks affect (expected) returns without affecting cashflows.

³In this case, we are able to characterize the equilibrium in closed form, even though prices are nonlinear functions of the state variables. In contrast, the model with intertemporal consumption choices is solved numerically.

Consequently, conditional on prices and aggregate wealth, any changes in noise traders' stock demand must be matched by changes in investors' aggregate bond demand. Under the traditional assumption of a bond in perfectly elastic supply, such changes in aggregate bond demand do not affect the interest rate; quantities, not prices, adjust. In contrast, with a imperfectly elastic (e.g., fixed) bond supply, the interest rate adjusts. As a result, the bond market (through the rate of interest) provides a signal that allows investors to learn about noise traders' stock demand, with the signal error originating from noise traders' bond demand.

For example, a high stock price is compatible with a strong stock demand emanating from both rational and noise traders. A high interest rate reveals further that rational investors' demand for bonds is weak, and hence that they—not noise traders—are the more likely source of the strong stock demand. Thus, the combination of a high stock price with a high interest rate signals a high stock payoff. This economic intuition extends to cases with intertemporal consumption choices, with the caveat that the bond-market signal then also contains information about the stock's payoff (through an income effect); but the precision of this information is low relative to that of information about the stock's noisy demand.

The second insight from our model is that the precision of the bond-market signal is *positively* correlated with the rate of interest—through the cross-market learning channel we just described. Indeed, when the interest rate is high (and so the bond price is low), the aggregate-resource condition—and, accordingly, the bond-market signal—is less sensitive to noise traders' bond demand (since it enters multiplied by the bond price). At the same time, a high interest rate signals a strong stock demand from rational investors (as explained above). This leads investors to revise upwards their beliefs about the stock's payoff, which raises the stock price, and, hence, renders the aggregate-resource condition more sensitive to noise traders' stock demand. With more variations in the unknown (stock noise) and less in the error (bond noise), the bond-market signal—and as a result the stock price—are more informative. This leads to a positive link between the interest rate and the informativeness of the stock price.⁴ In contrast, in standard REE models, the precision of both the stock-price signal and investors' posterior beliefs is independent

⁴To be precise, this property requires the stochastic part of noise traders' bond demand to contain a price-inelastic component—as stipulated in, e.g., preferred-habitat theories of the term structure and micro-founded later in the paper. Indeed, to model the noise traders' bond demand, we follow preferred-habitat theories of the term structure of interest rates (see, e.g., [Vayanos and Vila 2021](#)) and assume that their demand features both price-inelastic and price-elastic components. This is also consistent with our empirical finding of a positive relationship between stock price informativeness and interest rates as well as with bond supply (see [Figure 1](#) or [Table 1](#)).

of asset prices.

We then drive three implications of learning from the interest rate for allocative efficiency and for asset prices. First, the higher price informativeness allows firms to better differentiate between high-productivity and low-productivity states, and so to invest more efficiently. Consequently, allocative efficiency also correlates positively with the rate of interest. Second, because risk is reduced (thanks to higher price informativeness), the rate of interest also negatively correlates with the stock’s expected excess return and its return volatility. Third, the bond-stock correlation is stochastic and can change sign.

We also consider two extensions of our main framework featuring *additional signals*. Both not only confirm the key economic mechanism but also deliver new insights. The first extension, which allows for multiple stocks, shows that the bond-market signal induces a negative (cross-sectional) correlation between stocks’ excess returns (which declines in the interest rate)—despite fundamentals and noise trading being independent. The second extension, which includes money, demonstrates that, similar to the rate of interest, the rate of *inflation* provides information about noise traders’ stock demand (i.e., discount rate news). Moreover, just as the precision of the bond-market signal increases in the rate of interest, the precision of the money-market signal is increasing in the rate of inflation.

Finally, we consider the policy implications of the bond-market-learning channel. We examine how governments’ and central banks’ interventions in the bond market affect informational and allocative efficiency in the stock market. For example, we show that efficiency worsens as bond purchases intensify (e.g., through QE programs) whereas it improves as they become more transparent (e.g., by improving disclosure). Similarly, limits on interest rates (such as the “zero lower bound” or restrictions on their fluctuations) harm efficiency.

Overall, these results demonstrate that the interest rate affects the stock market and the real economy not only through its traditional impact on discount rates but also through its impact on the information environment. Indeed, our work highlights important adverse consequences of the decline in long-term interest rates in recent decades (irrespective of underlying cause).

Several implications of the model are consistent with broad features of the data. For instance, our model predicts that stock-price informativeness increases in the real interest rate (and in bond and money supplies), in line with our empirical investigation. Related, the model predicts that allocative efficiency should be high (low) in high (low) interest-rate environments. This prediction is consistent with the empirical evidence presented by [Gopinath et al. \(2017\)](#), who document a simultaneous decline in real interest rates

and capital allocation efficiency in southern European countries. Moreover, in the model, periods of low interest rates are associated with an increase in the mean and variance of excess returns, and in stock-return comovement. Combined with the cyclicalities of interest rates observed in the data, these results imply that excess returns as well as the volatility and comovement of stock returns, are all countercyclical, as in the data. Finally, also consistent with the data (see, among others, [Fleming, Kirby, and Ostdiek 2003](#)), our model predicts that the bond-stock correlation can switch sign—without relying on counterfactual changes in inflation expectations (cf. [Duffee 2021](#)). More work is needed to ascertain whether these associations are actually causal or mere correlations.

Related literature. The paper spans several strands of the literature. Foremost, it builds on the noisy REE literature pioneered by [Grossman and Stiglitz \(1980\)](#) and [Hellwig \(1980\)](#). Our main contribution to this literature is to endogenize the rate of interest. We show that the interest rate contains valuable information about a stock’s noisy demand and work out how investors use this information to update their beliefs about a stock’s payoff. We are not aware of any other work in which *both* stock prices and the interest rate reveal information.⁵ That price informativeness and investors’ posterior precision endogenously vary with the rate of interest and, hence, with the business cycle further distinguishes our model from other noisy REE models. This property also connects our work to that of [Kacperczyk, van Nieuwerburgh, and Veldkamp \(2016\)](#), who analyze how investors’ knowledge depends on the state of the economy. Their mechanism differs from ours in that this dependence on the state of the economy stems from variations in risk and in the price of risk ([Kacperczyk, van Nieuwerburgh, and Veldkamp 2016](#)) versus variations in the interest rate (our model).

Our paper further relates to three substreams of the noisy REE literature. The first studies economies with multiple assets (see, e.g., [Admati 1985](#), [van Nieuwerburgh and Veldkamp 2009, 2010](#), [Biais, Bossaerts, and Spatt 2010](#), [Kacperczyk, van Nieuwerburgh, and Veldkamp 2016](#)). Though our main model features two assets with informative prices, it distinctly differs from these models in that our second asset is *risk-free*. Indeed, we show that the risk-free asset reveals information about the stock despite their payoffs and noisy demands being *uncorrelated*. This is in sharp contrast to [Admati \(1985\)](#) and

⁵[Detemple \(2002\)](#) also proposes an REE model in which the interest rate is endogenous. However, in his setting, information is revealed only through the interest rate (whereas the stock price does not reveal any information), and, moreover, this information is exclusively about cashflows (rather than to discount rates as in our model). Indeed, the economic mechanism and insights differ substantially; for example, the (positive) link that we highlight between the interest rate and stock-price informativeness is absent from his framework.

the work that followed, in which, absent (exogenous) cross-asset correlations, nothing is to be learned from one asset about another. In addition, in our extension to multiple stocks, bond market clearing endogenously generates a (negative) correlation between stocks' returns (which is also absent from [Admati 1985](#)). Note also that in these models price informativeness is deterministic, whereas it is stochastic in our framework. Second, through its emphasis on information about the stock's noisy demand, our work is also related to [Watanabe \(2008\)](#), [Ganguli and Yang \(2009\)](#), [Manzano and Vives \(2011\)](#), [Farboodi and Veldkamp \(2019\)](#), and [Yang and Zhu \(2019\)](#). In these papers, investors receive a private and exogenous signal (which they either purchase or are endowed with) about the stock supply. In contrast, in our setup, the supply signal (also referred to as order flow information in the literature) is public and endogenous—generated by the bond market. Third, our paper is part of the substream of the literature that seeks to generalize noisy REE models and explore their robustness to assumptions (see, e.g., [Barlevy and Veronesi 2000, 2003](#), [Peress 2004](#), [Breon-Drish 2015](#), [Banerjee and Green 2015](#), [Albagli, Hellwig, and Tsyvinski 2015](#)). Our contribution is to endogenize the interest rate in an otherwise standard noisy REE model and identify what features survive or differ.

Finally, our paper relates to the literature studying the importance of an endogenous rate of interest in asset pricing models under *symmetric* information. [Lowenstein and Willard \(2006\)](#) highlight that the assumption of a riskless asset in perfectly elastic supply can yield misleading conclusions. Our work differs from their paper because of the presence of private information and our focus on price informativeness. Moreover, we find that the main conclusions of the traditional noisy REE literature are robust to endogenizing the interest rate. Instead, we illustrate that new mechanisms arise when the bond market clears under a fixed bond supply.

The remainder of the paper is organized as follows. Section [II](#) presents the novel empirical findings motivating our theoretical analysis. Section [III](#) introduces our main economic framework. Section [IV](#) discusses, in a tractable version of the model, the economic mechanism through which investors learn from the interest rate. In Section [V](#), we then study the full model and relate the characteristics of the bond market to equilibrium outcomes. Sections [VI](#) and [VII](#), respectively, explore extensions with additional price signals and policy implications. Finally, Section [VIII](#) concludes.

II. Empirical Patterns in Price Informativeness

In this section, we offer novel empirical evidence on the relation between the informativeness of stock prices and the rate of interest. We document that, in the data, stock-price informativeness is associated with long-term interest rates.

II.A. Data and Estimation Procedures

Our analysis focuses on the U.S. market over the period from 1962 to 2017.

Price Informativeness: We measure the informativeness of stock prices using the proxy developed by [Bai, Philippon, and Savov \(2016\)](#) which captures the extent to which firms' current stock prices reflect their future cashflows. Specifically, in each year, we run the following cross-sectional regression of year- $t+h$ earnings on year- t stock prices:

$$\frac{E_{j,t+h}}{A_{j,t}} = a_{t,h} + b_{t,h} \log \left(\frac{M_{j,t}}{A_{j,t}} \right) + c_{t,h} X_{j,t} + \epsilon_{j,t,h}, \quad (1)$$

where h denotes the forecasting horizon; $E_{j,t+h}/A_{j,t}$ denotes firm j 's earnings before interest and taxes (EBIT) in year $t+h$ scaled by year- t total assets; $M_{j,t}/A_{j,t}$ denotes firm j 's market capitalization (i.e., stock price times the number of shares outstanding) in year t scaled by year- t total assets; and $X_{j,t}$ denotes a set of firm-level controls, namely, current earnings, $E_{j,t}/A_{j,t}$, and industry fixed effects (one-digit SIC codes).⁶

Intuitively, the coefficient $b_{t,h}$ reflects how closely current stock prices track future earnings and, hence, how much fundamental information is capitalized in stock prices. Price informativeness at horizon h , $PI_{t,h}$, is then measured as the coefficient estimate $b_{t,h}$ multiplied by the year- t cross-sectional standard deviation of (scaled) stock prices:

$$PI_{t,h} = b_{t,h} \times \sigma_t \left(\log \left(\frac{M_{j,t}}{A_{j,t}} \right) \right). \quad (2)$$

As discussed in [Bai, Philippon, and Savov \(2016\)](#), $PI_{t,h}^2$ captures the variance of the predictable component of firms' payoffs, F_j , given its stock price: $\text{Var}(\mathbb{E}[F_j | P_j])$. Hence, $PI_{t,h}$ serves as a natural proxy for forecasting price efficiency.

⁶To align price informativeness with bond market characteristics, we sample stock prices at the end of the U.S. government's fiscal year (either June or September). For each firm, we measure accounting variables at the end of the previous fiscal year—typically December—to ensure that the information is readily available to market participants. We adjust earnings using the gross domestic product (GDP) deflator from the Bureau of Economic Analysis (BEA).

We obtain stock price data from CRSP and accounting data from Compustat. Like [Bai, Philippon, and Savov \(2016\)](#), we focus on S&P 500 nonfinancial firms whose characteristics have remained remarkably stable over time. Moreover, we concentrate on forecasting horizons (h) of 3 and 5 years; horizons that, from a capital allocation perspective, are most important (see, e.g., the time-to-build literature, in particular, [Koeva 2000](#)) and for which prices are particularly useful in predicting earnings (as reported in [Bai, Philippon, and Savov 2016](#)).

Interest Rates: Our bond-market measures closely follow those used by [Krishnamurthy and Vissing-Jorgensen \(2012\)](#). U.S. real interest rates are obtained by deducting expected inflation from long-term nominal rates. The nominal rate on long-maturity Treasury bonds is measured as the average yield on government bonds with a maturity of 10 years or longer (up to 1999) and the 20-year Treasury constant-maturity rate (from 2000 on), both of which are obtained from the Federal Reserve’s FRED database. Expected inflation is estimated using a simple random-walk model (applied to the Consumer Price Index of the BEA).⁷

We measure the supply of U.S. Treasuries as the ratio of the market value of publicly held U.S. government debt to GDP. For that purpose, we adjust the book value of U.S. government debt (obtained from the annual Treasury Bulletin) by the Treasury debt market price index provided by the Dallas Fed.⁸ Furthermore, we proxy for the demand for U.S. Treasuries by the Federal Reserve banks’ holdings of Treasury securities which we also instrument using their holdings of mortgage-backed securities (MBS). Government debt, Treasury holdings, and, accordingly, GDP are all measured at the end of the government’s fiscal year (i.e., the end of June up to 1976 and the end of September from 1977 onward).

Control Variables: We estimate stock market and cashflow volatility as the annualized standard deviation of daily S&P 500 returns over the past 12 months and the cross-sectional standard deviation of firms’ (scaled) earnings, respectively.

Table IA1 in the Online Appendix reports summary statistics for all variables.

⁷The random-walk model delivers the best out-of-sample performance for predicting inflation. Our findings are robust to the use of alternative models for expected inflation, namely, AR(1) and ARMA(1,1) models.

⁸We obtain similar results with the debt-to-GDP series of [Krishnamurthy and Vissing-Jorgensen \(2012\)](#).

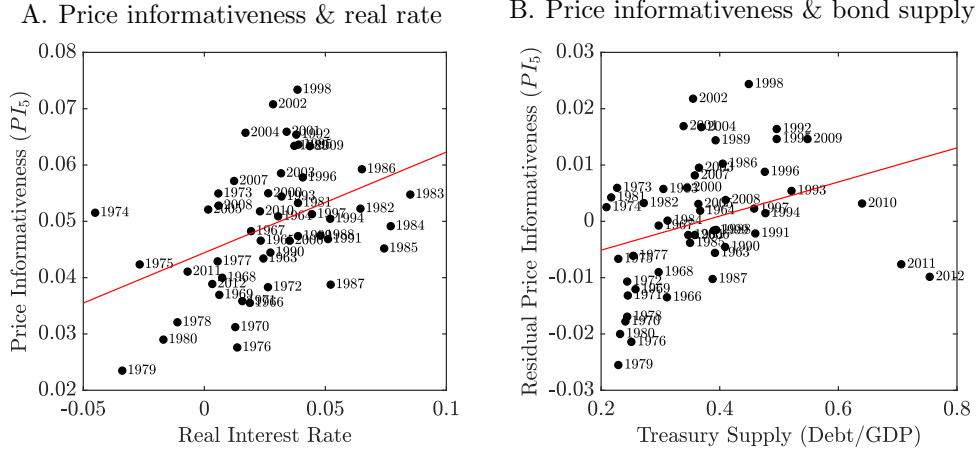


FIGURE 2. EMPIRICAL PATTERNS IN STOCK-PRICE INFORMATIVENESS

Notes: The panels plot stock price informativeness against the real interest rate (Panel A) and the debt-to-GDP ratio (Panel B). The sample consists of annual observations from 1963 to 2012. Residual price informativeness (Panel B) is measured as the residuals of a univariate regression of price informativeness on the Federal Reserve Banks' MBS holdings. The solid line in all graphs represents the fitted values of a univariate regression of the y -axis variable on the x -axis variable.

II.B. Price Informativeness and the Rate of Interest

We start by regressing five-year stock-price informativeness, PI_5 , on the real interest rate. The resulting estimate is positive (0.179) and highly significant (t -statistic of 2.67), as Panel A of Figure 2 illustrates. In terms of economic magnitude, a one-standard-deviation (SD) increase in the real interest rate leads to a 0.42-SD increase in price informativeness.⁹

While this finding clearly points towards a positive correlation between price informativeness and the interest rate, bear in mind that the rate of interest is endogenous; that is, it is determined in equilibrium jointly with other quantities, including price informativeness. Hence, in our next analysis, we instrument for variations in the real interest rate using our proxies for Treasury supply and demand. Indeed, it seems implausible that the government chooses its debt level or that Federal Reserve Banks choose their Treasury or MBS holdings in accordance with the informativeness of stock prices.

Table 1 reports the results of our regression analyses. The dependent variable in each regression is price informativeness (typically PI_5) and the primary explanatory variables

⁹While this positive link might, at first glance, seem inconsistent with the upward trend in price informativeness documented by Bai, Philippon, and Savov (2016) and the concurrent downward trend in interest rates, note that our estimation is based on annual variations in price informativeness and real interest rates, rather than longer-term trends. For instance, as Panel A of Figure 2 illustrates, price informativeness has been particularly low during the early 2010s and the mid-to-late 1970s/early 1980s—two periods known for their low real rates.

TABLE 1—STOCK-PRICE INFORMATIVENESS AND THE BOND MARKET

	Base	1963- 2009	5-year periods	FED: Treasury	Lagged variables	Volatility Controls		PI_3
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Debt/GDP	0.060 [3.13]	0.074 [4.97]	0.079 [3.24]	0.048 [3.40]	0.066 [4.09]	0.063 [3.21]	0.061 [3.34]	0.033 [1.98]
FED Hold./GDP	-0.331 [-2.52]		-0.452 [-2.27]	-0.364 [-3.38]	-0.421 [-5.45]	-0.369 [-2.55]	-0.359 [-3.21]	-0.248 [-2.36]
S&P500 Vola.						0.028 [0.81]		0.037 [1.03]
Cashflow Vola.							0.664 [3.12]	0.506 [2.45]
R^2	0.211	0.336	0.600	0.228	0.260	0.226	0.350	0.235
Observations	50	46	10	50	50	50	50	50

Note. The table reports results of regressions relating stock price informativeness to Treasury-bond supply and demand. The dependent variable is 5-year price informativeness, PI_5 , (except in Column 8 which is based on 3-year price informativeness, PI_3). *Debt/GDP* is the ratio of the market value of Treasury debt held by the public to U.S. GDP. *FED Hold./GDP* is the ratio of the Federal Reserve banks' holdings of MBS (or Treasury in Column 4) divided by U.S. GDP. *S&P500 Vola.* and *Cashflow Vola.* are measures of volatility of, respectively, the S&P500 returns and firms' earnings. Regressions are estimated using OLS and standard errors are adjusted for serial correlation using the Newey-West procedure with five lags. We report t -statistics in brackets.

are the Treasury-bond supply and demand. The regressions in Table 1 are estimated using ordinary least squares, with standard errors adjusted for serial correlation using the Newey-West procedure with five lags.¹⁰

The baseline regression in Column 1 confirms the link between price informativeness and the rate of interest, with the bond supply positively (t -statistic of 3.13) and the bond demand negatively (t -statistic of -2.52) correlating with stock-price informativeness. Quantitatively, all else equal, a one-SD increase in the debt-to-GDP ratio (from its mean value of 0.383 to 0.494) increases price informativeness by 15% (0.64 SD). Panel B of Figure 2, which plots residual price informativeness (i.e., the residuals of a univariate regression of price informativeness on Treasury demand) against the Treasury supply, illustrates this finding.

The remainder of Table 2 confirms that our findings hold up to a series of robustness checks. Column 2 focuses on the period from 1962 to 2009, over which Treasury demand was constant and so does not need to be controlled for. Column 3 exploits only low-frequency variations in the series; that is, it reports the results of a regression

¹⁰Our choice of lags is based on two considerations. First, price informativeness is measured by overlapping regressions, with a maximum overlap of five years. Second, the optimal lag-selection-procedure of Newey and West (1994) recommends lags between three and five years. Our results are robust to alternative specifications.

of (nonoverlapping) five-year averages of the variables (i.e., a total of 10 data points). Column 4 uses the FED’s Treasury holdings (instead of their MBS holdings) to control for Treasury demand. Column 5 lags bond supply and demand. Columns 6 and 7 control for stock market and cashflow volatility, respectively. Finally, Column 8 uses the price-informativeness measure, PI_3 , based on a three-year forecasting horizon.

Taken together, these results provide robust evidence that price informativeness correlates with long-term real interest rates, a finding which poses a substantial challenge to traditional information choice models and motivates our subsequent theoretical analysis.

III. An REE Model with Bond Market Clearing

In this section, we introduce our economic framework. The framework differs from traditional competitive REE models, such as those of [Grossman and Stiglitz \(1980\)](#), [Hellwig \(1980\)](#), and [Verrecchia \(1982\)](#), along three key dimensions. First, the bond market clears, and, hence, the rate of interest is endogenously determined. Second, investors learn not only from their private signals and the stock price but also from the interest rate. Third, agents consume not only in the final period but also in the initial period (with the interest rate reflecting their consumption smoothing motives). Moreover, to illustrate the implications for allocative efficiency, we endogenize firms’ real-investment decisions and, thus, output. In the following, we describe the details of the model.

Information Structure and Timing

We consider a two-period model. [Figure 3](#) illustrates the sequence of events. In period 1, investors observe their private signals, the stock price, and the interest rate. Based on this information, they set up their portfolio and choose period-1 (“initial”) consumption. In addition, a representative firm chooses its real investment, conditional on asset prices. Finally, asset prices clear financial markets. In period 2, productivity and output are realized, and investors consume the proceeds from their investments (“terminal” consumption).

Investment Opportunities

Two financial securities are traded in competitive markets: a riskless asset (the “bond”) and a risky asset (the “stock”). The bond has a payoff of one in period 2, with a gross

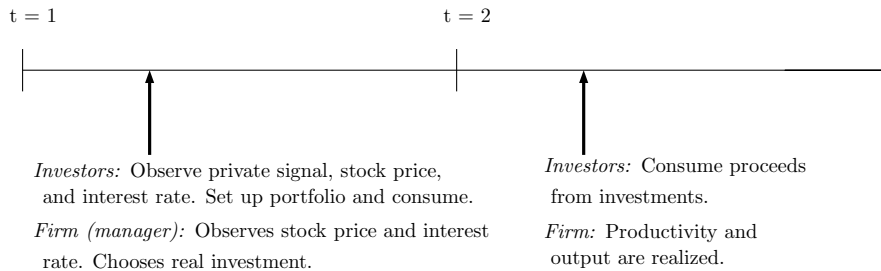


FIGURE 3. SEQUENCE OF EVENTS

Notes: The figure illustrates the sequence of the events.

rate of interest R_f , or, equivalently, a price of $1/R_f$.^{11,12} The stock is a claim to the representative firm's endogenous output F (the “fundamental”), which is only observable in period 2. Its price is denoted by P . The stock and the bond are in finite supply, denoted by $\bar{X}^S$ and $\bar{X}^B$, respectively.

Output

Output is produced by a representative firm that employs a linear (“ZK”) production technology and is endowed with assets in place K_1 . Its fundamental value, v , is modeled as in standard q -theory:

$$v(z, I) \equiv \underbrace{(K_1 - I)}_{\equiv F_1} + \underbrace{(1+z) \left((1-\delta) K_1 + I \right) - \frac{\kappa}{2 K_1} I^2}_{\equiv F}. \quad (3)$$

Specifically, with period-1 productivity being normalized to one, initial-period output, F_1 , is simply given by assets-in-place K_1 less investment I . As is standard in such models, investment, I , can be positive (representing capital expenditures) or negative (representing an asset sale). Period-2 output, F , is given by the product of period-2 productivity, $1 + z$, and available capital (assets-in-place K_1 depreciated at rate δ , plus investment I), minus quadratic adjustment costs $(\kappa/2K_1)I^2$ (with $\kappa \geq 0$). Period-2 net productivity, z , is stochastic and normally distributed with mean zero and precision τ_z :

¹¹In our setting, the consumption good serves as the numéraire, and, hence, all prices (and payoffs) are denominated in units of the good. This contrasts with traditional REE models, such as those of Grossman and Stiglitz (1980), Hellwig (1980), and Verrecchia (1982), in which the exogenous riskless bond serves as the numéraire.

¹²The model is immune to how we denominate the bond price; e.g., one could also set its price to 1 and its payoff to R_f .

$z \sim \mathcal{N}(0, 1/\tau_z)$.

For simplicity, we assume that the firm (manager) has no private information about productivity, z , but learns about productivity from stock and bond prices.¹³ This creates a *feedback effect* from financial markets to real investment decisions.¹⁴

Investors

There exists a continuum of atomless investors with unit mass. In period 1, each investor i receives a private signal about productivity: $s_i = z + \varepsilon_i$, where ε_i is *i.i.d.* normally distributed with mean zero and precision τ_ε . Investors have constant absolute risk aversion (CARA) preferences over initial and terminal consumption, $C_{i,1}$ and $C_{i,2}$:

$$\mathcal{U}_i(C_{i,1}, C_{i,2}) = -\frac{1}{\rho} \exp(-\rho C_{i,1}) + \beta \mathbb{E} \left[-\frac{1}{\rho} \exp(-\rho C_{i,2}) \mid \mathcal{F}_i \right], \quad (4)$$

where ρ denotes absolute risk aversion; $\beta \in (0, 1]$ denotes the rate of time preference; and $\mathcal{F}_i = \{s_i, P, R_f\}$ describes investor i 's time-1 information set. Investors are each initially endowed with $W_{i,1}$ units of the good (or, equivalently, of a maturing bond).¹⁵

Noise Traders

Noise (liquidity) traders operate in *both* the bond and the stock market. In particular, we assume, in addition to the usual stock market noise, a noisy bond demand; this assumption prevents the bond and stock prices from being *jointly* perfectly revealing.

Noise traders' behavior is not explicitly modeled; instead, their demands are given by exogenous random variables. Specifically, following the literature, noise traders' demand for the stock equals $u^S \sim \mathcal{N}(0, 1/\tau_{u^S})$. To model the noise traders' bond demand, we follow preferred-habitat theories of the term structure of interest rates (see, e.g., [Vayanos and Vila 2021](#)) and assume that their demand features both price-inelastic and price-elastic components.¹⁶ Specifically, noise traders' demand for the bond equals $\alpha R_f + u^B$, where $u^B \sim \mathcal{N}(0, 1/\tau_{u^B})$. z , u^S , and u^B are assumed uncorrelated. This correlation

¹³In particular, in our single-stock economy, the firm represents the entire productive sector, so z can be interpreted as *aggregate* productivity, about which the manager has plausibly no private information.

¹⁴[Bond, Edmans, and Goldstein \(2012\)](#) survey the literature on feedback effects. For more recent contributions, see [Foucault and Frésard \(2014\)](#), [Edmans, Goldstein, and Jiang \(2015\)](#), [Goldstein and Yang \(2017\)](#), and [Dessaint et al. \(2018\)](#).

¹⁵Our main findings hold if investors have preferences other than CARA or are endowed with shares of the stock (see the Online Appendix).

¹⁶Appendix A presents a micro-foundation for this type of noise.

structure highlights that, in equilibrium, the bond price reveals information about the stock even though its payoff and noise are uncorrelated with those of the stock.

Equilibrium Definition

Investor i aims to maximize expected utility (4) subject to the following budget constraints:

$$C_{i,1} + X_i^S P + X_i^B R_f^{-1} = W_{i,1}, \quad \text{and} \quad C_{i,2} = X_i^S F + X_i^B, \quad (5)$$

where X_i^S and X_i^B denote the number of shares of the stock and the bond held by the investor, respectively. The objective of the manager is to maximize the expected firm value.

A rational expectations equilibrium is defined by consumption choices $\{C_{i,1}, C_{i,2}\}$, portfolio choices $\{X_i^S, X_i^B\}$, a real investment choice I , and asset prices $\{P, R_f\}$ such that

1. $\{C_{i,1}, C_{i,2}\}$ and $\{X_i^S, X_i^B\}$ maximize investor i 's expected utility (4) subject to the budget constraints in (5), taking prices P and R_f as given,
2. I maximizes the expected firm value $\mathbb{E}[v(z, I) \mid R_f, P]$,
3. the investors' and the manager's expectations are rational,
4. aggregate demand equals aggregate supply in the bond and the stock markets.¹⁷

$$\int_0^1 X_i^S di + u^S = \bar{X}^S, \quad \text{and} \quad \int_0^1 X_i^B di + \alpha R_f + u^B = \bar{X}^B. \quad (6)$$

Importantly, *both* asset prices play a dual role in equilibrium: each price not only clears its respective market but also aggregates and transmits investors' private information.

IV. The Economic Mechanism

We now first illustrate *how* and *what type of information* investors learn from the bond market. To do so, we use a simplified version of our model that provides the key economic intuition thanks to closed-form solutions. It deviates from the framework described in the preceding section only in that investors consume exclusively on the terminal date.

¹⁷By Walras' law, clearing the bond and stock markets guarantees that the goods market clears.

Moreover, to facilitate the exposition of the economic mechanism, we abstract from real investment and treat the stock's payoffs, F , as exogenous: $F \sim \mathcal{N}(\mu_F, 1/\tau_F)$. Hence, the model essentially reduces to Hellwig's (1980) but with bond market clearing.

IV.A. Equilibrium

Because of learning from the interest rate, asset prices are *nonlinear* functions of the state variables—in stark contrast to traditional frameworks. However, by conjecturing the functional form of the market-clearing conditions (which remain linear), instead of stipulating the functional form of the interest rate and the stock price (which are not linear), we can characterize the equilibrium in closed form, as stated in the following theorem:

Theorem 1. *There exists a unique (conditionally linear) rational expectations equilibrium. The equilibrium asset prices are characterized by*

$$R_f = \frac{\bar{X}^B - u^B}{\bar{W}_1 + \alpha - (\bar{X}^S - u^S) P}; \quad \text{and} \quad (7)$$

$$R_f P = \frac{1}{\tau} \left(-\rho \bar{X}^S + \tau_F \mu_F - \frac{\rho PI}{\tau_\varepsilon} \mu_{u^S|s^B} \right) + \frac{\tau_\varepsilon + PI}{\tau} \left(F + \frac{\rho}{\tau_\varepsilon} u^S \right), \quad (8)$$

$$\text{with } \bar{W}_1 \equiv \int_0^1 W_{i,1} di, \quad PI \equiv (\tau_\varepsilon/\rho)^2 \tau_{u^S|s^B}, \quad \tau \equiv \tau_F + \tau_\varepsilon + PI, \quad \text{and}$$

$\mu_{u^S|s^B}$ and $\tau_{u^S|s^B}$ being defined in Equations (12) and (13) below.

Investor i 's optimal stock and bond holdings equal

$$X_i^S = \frac{\mathbb{E}[F | \mathcal{F}_i] - R_f P}{\rho \text{Var}(F | \mathcal{F}_i)} \quad \text{and} \quad X_i^B = R_f (W_{i,1} - X_i^S P). \quad (9)$$

The demand for the stock follows the standard mean-variance portfolio rule, and the bond expenditures are simply given by the remaining wealth.

The equilibrium interest rate, R_f , is a function of the realizations of noise traders' stock and bond demands and, thus, is stochastic.¹⁸ As expected, it is increasing in the bond supply, $\bar{X}^B$, and declining in the deterministic and stochastic components of the

¹⁸ The gross interest rate, R_f , can be negative in this illustrative framework. This does not, however, lead to arbitrage opportunities. Indeed, negative rates are caused by investors deriving utility from terminal consumption only (i.e., they do not consume in period 1), and, thus—in contrast to the economy analyzed in Section V—the interest rate is entirely determined by budget considerations and not by marginal utilities.

bond demand, α and u^B . In addition, the interest rate declines in investors' aggregate wealth, $\bar{W}_1$, and increases in their stock expenditures, $(\bar{X}^S - u^S) P$ (as higher stock expenditures imply less wealth is allocated to the bond market; and so a lower bond price).

The equilibrium price ratio, $R_f P$, has the familiar structure of, e.g., [Hellwig \(1980\)](#), with one critical difference: it features the mean and precision of the noisy stock demand, $\mu_{u^S|s^B}$ and $\tau_{u^S|s^B}$, *conditional* on the bond-market signal, instead of its prior mean and precision. This difference arises because investors use the information revealed by the bond market to update their beliefs about noise traders' stock demand (see next section).

IV.B. Learning from the Bond Market

In the absence of initial consumption, investors' period-1 budget constraints, combined with market clearing, imply that stock and bond expenditures (modulo the deterministic component of bond demand, α) sum up to aggregate wealth or, formally:

$$(\bar{X}^S - u^S) P + (\bar{X}^B - u^B) R_f^{-1} - \alpha = \bar{W}_1. \quad (10)$$

This aggregate-resource constraint—which ties together the stock and bond demands of the noise traders—is the equilibrium condition that determines the interest rate. It also provides investors with a signal, s^B , of the form: Positive signal -> Low P/B -> High PI -> good news more informative

$$s^B \equiv -\frac{R_f (\bar{W}_1 + \alpha) - \bar{X}^S R_f P - \bar{X}^B}{R_f P} = u^S + \frac{u^B}{R_f P}, \quad (11)$$

where s^B represents the aggregate wealth of noise traders denominated in shares of the stock. In words, *the bond market provides information about the unobservable stock demand, u^S , with the bond demand, u^B , serving as noise*. The following lemma describes the resultant conditional distribution of the noisy stock demand.

Lemma 1. *The distribution of the noisy stock demand, u^S , conditional on the bond-market signal s^B is characterized by*

$$\mu_{u^S|s^B} \equiv \mathbb{E}[u^S | s^B] = \frac{(P R_f)^2 \tau_{u^B}}{\tau_{u^S|s^B}} \frac{\bar{X}^S P + \bar{X}^B R_f^{-1} - (\bar{W}_1 + \alpha)}{P}; \text{ and} \quad (12)$$

$$\tau_{u^S|s^B} \equiv \text{Var}(u^S | s^B)^{-1} = \tau_{u^S} + (P R_f)^2 \tau_{u^B}. \quad (13)$$

Investors combine their priors with the signal provided by the bond market to update

their beliefs about the noisy stock demand. The conditional mean, $\mu_{u^S|s^B}$, is the precision-weighted average of the prior mean (equal to zero) and the bond signal in (11). Similarly, the conditional precision, $\tau_{u^S|s^B}$, is the sum of the prior precision (τ_{u^S}) and the precision of the bond-market signal ($P^2 R_f^2 \tau_{u^B}$).

Note that the conditional precision, $\tau_{u^S|s^B}$, is increasing in the price ratio, $R_f P$.¹⁹ Intuitively, because the bond signal s^B stems from investors' aggregate-resource constraint (10), a higher price ratio implies that the signal assigns a higher weight to noise traders' stock demand u^S relative to their bond demand u^B , which makes the signal more informative about the former. For example, suppose the stock price and the interest rate are both high; then practically all variations in investors' expenditures (left-hand-side of Equation (10)) originate from stock expenditures, making it straightforward for investors to infer the noisy stock demand u^S .

The following lemma characterizes investors' posterior beliefs regarding the payoff F .

Lemma 2. *Investor i 's posterior beliefs regarding the payoff F are given by:*

$$\mathbb{E}[F | \mathcal{F}_i] = \frac{1}{\tau} \left(\tau_F \mu_F + \tau_\varepsilon s_i + \frac{PI}{\tau_\varepsilon + PI} (\tau R_f P - \tau_F \mu_F - \rho (\bar{X}^S + \mu_{u^S|s^B})) \right), \quad (14)$$

$$\text{Var}(F | \mathcal{F}_i)^{-1} = \tau_F + \tau_\varepsilon + (\tau_\varepsilon / \rho)^2 \tau_{u^S|s^B}. \quad (15)$$

The payoff's posterior mean and precision, $\mathbb{E}[F | \mathcal{F}_i]$ and $\text{Var}(F | \mathcal{F}_i)^{-1}$, share many similarities with the expressions in Hellwig (1980). The novel feature of our framework is that they are related to the interest rate.

First, the posterior mean (14) is a precision-weighted sum of investors' signals (prior, private signal, and public stock price), with the interest rate entering through the stock price signal. For instance, a high stock price reveals a high aggregate stock demand but leaves unclear who is driving the demand. Information from the bond market helps assess whether the strong stock demand emanates mostly from rational traders or from noise traders. Specifically, suppose investors also observe a high interest rate. Then they infer that rational investors' bond expenditures are rather low; hence, keeping wealth fixed, they deduce that the high stock demand most likely stems from rational investors or, equivalently, that noise traders' stock demand is likely to be low (i.e., $\mu_{u^S|s^B}$ in Equation (12) is low). Given this expectation, they conclude—from the high stock price—that the stock's payoff is likely to be high (i.e., $\mathbb{E}[F | \mathcal{F}_i]$ in Equation (14) is high).

¹⁹There exists a *two-way relationship* between the price ratio and the precision of information: the higher the price ratio, the more precise is information (as Equation (13) shows); conversely, the more precise information, the higher the price ratio (as Equation (8) shows).

Second, the posterior precision is the sum of three components: (1) the precision of the investors' prior beliefs τ_F , (2) the precision of their private signal τ_ε , and (3) the precision of the stock price signal $PI \equiv (\tau_\varepsilon/\rho)^2 \tau_{us|s^B}$. Again, the interest rate enters through the stock price signal. Its precision depends on the precision of investors' information about the noisy stock demand. In contrast to Hellwig (1980), where it is equal to investors' prior precision, that precision is not exogenous but equal to investors' precision *conditional* on the bond-market signal, $\tau_{us|s^B}$ (see Equation (13)). This observation has three important implications which distinguish our model from traditional noisy REE models. First, investors' posterior precision is higher than in Hellwig (1980), thanks to the information on the noisy stock demand obtained from the bond market. Second, the stock signal provides information about the stock's payoff even if the variance of noise trading in the stock market is infinite ($\tau_{us} = 0$) because, conditional on the bond-market signal, that variance is finite ($\tau_{us|s^B} > 0$). Indeed, in this case, all learning stems from the bond market. Third, price informativeness depends on asset prices and, hence, is stochastic and *ex ante* unknown.

IV.C. Price Informativeness and the Rate of Interest

We next examine the link between stock-price informativeness and the interest rate. As Figure 4 illustrates, price informativeness and the interest rate are endogenously correlated in our model—as they are in the data (cf. Section II).²⁰

To understand the underlying mechanism, let's focus on shocks to the bond demand, u^B , keeping fixed the payoff, F , and the noisy stock demand, u^S . For concreteness, consider a low realisation of the noisy bond demand, which implies a low total (i.e., investors' plus noise traders') demand for the bond.

In the absence of bond-market learning, the bond-market shock affects the bond price but not stock-price informativeness. Indeed, as the dashed lines in Figure 5 illustrate, the low aggregate bond demand leads to a high rate of interest (Panel A) but, because the two markets are informationally disconnected, it affects neither investors' beliefs regarding the stock's payoff (Panel B) nor stock-price informativeness (Panel D). Notably, while the high rate of interest depresses the stock price through a standard discounting effect, the price ratio $R_f P$ (i.e., the undiscounted stock price) is unaffected (Panel C).²¹

²⁰The graphs in this section are based on the following parameter values: $\rho = 4$, $\mu_F = 1$, $\tau_F = 2.5^2$, $\tau_\varepsilon = 1^2$, $\bar{X}^S = 1$, $\tau_{us} = 5^2$, $\bar{X}^B = 1$, $\tau_{u^B} = 10^2$, $\alpha = 0$, and $W_{i,1} = 1.5$.

²¹Indeed, with market clearing in the bond market but *without* learning from the bond market, the expression for the equilibrium price ratio, $R_f P$, coincides with that in Hellwig (1980). However, the

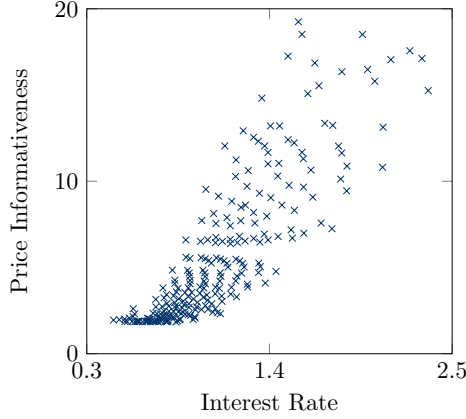


FIGURE 4. PRICE INFORMATIVENESS VS. INTEREST RATE (NO EARLY CONSUMPTION)

Notes: The figure plots price informativeness, PI , as a function of the rate of interest, R_f —in the model *without initial consumption*. Price informativeness is measured—state-by-state—as the precision of investors’ posterior beliefs in excess of prior precision, τ_F , and of the precision of their private signal, τ_ε . The graph is based on the following parameter values: $\rho = 4$, $\mu_F = 1$, $\tau_F = 2.5^2$, $\tau_\varepsilon = 1^2$, $\bar{X}^S = 1$, $\tau_{u^S} = 5^2$, $\bar{X}^B = 1$, $\tau_{u^B} = 10^2$, $\alpha = 0$, and $W_{i,1} = 1.5$.

Now suppose that investors learn from the bond market, as illustrated by the solid lines in Figure 5. As before, a low bond demand leads to a high rate of interest (Panel A). But there is now a cross-market learning effect: in interpreting the high interest rate, investors partly assign the low total bond demand to a strong *stock* demand from rational investors. Indeed, in light of the aggregate-resource constraint (Equation (10)), high stock expenditures imply, keeping wealth fixed, low bond expenditures by rational investors. This strong stock demand, in turn, leads investors to revise upwards their beliefs about the stock’s payoff (Panel B). As a result, the standard discounting effect of a higher interest rate is partially offset by investors’ more optimistic beliefs about the stock’s payoff. This pushes up the price ratio $R_f P$ (i.e., the undiscounted stock price), as shown in Panel C.

As discussed in Section IV.B, an increase in the price ratio, $R_f P$, in turn, renders the bond-market signal more informative about the noisy stock demand (because it attenuates the error created by bond noise) and, hence, increases stock-price informativeness. Thus, learning from the bond market creates a link from bond-market shocks, u^B , to price informativeness (Panel D). This, together with the natural link from bond-market shocks to the rate of interest (Panel A), creates an endogenous (positive) correlation between the rate of interest and stock-price informativeness.

interest rate remains stochastic, so that the equilibrium is not identical to Hellwig’s.

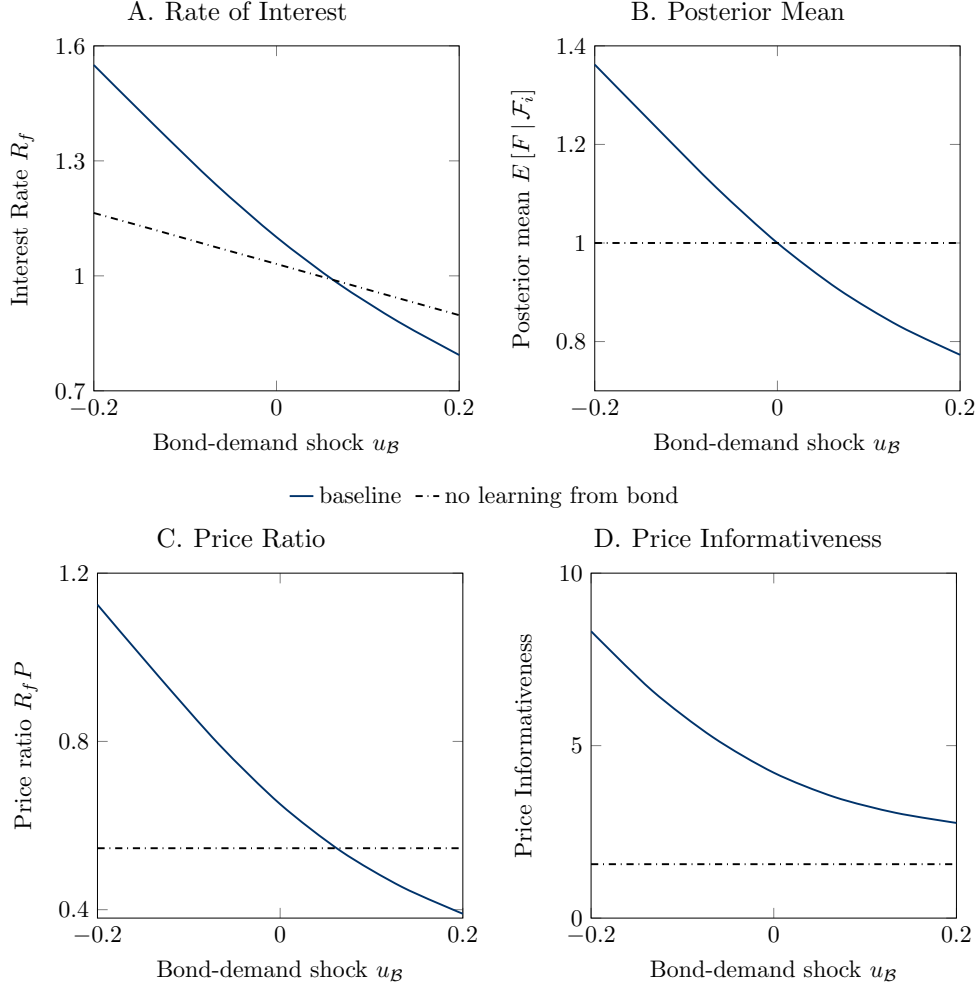


FIGURE 5. IMPACT OF BOND-MARKET DEMAND SHOCKS (NO EARLY CONSUMPTION)

Notes: The figure illustrates how bond-market demand shocks, u^B , affect the equilibrium—in the model *without initial consumption*. Specifically, Panels A to D, respectively, plot the rate of interest R_f , investors' posterior mean $\mathbb{E}[F | \mathcal{F}_i]$, the price ratio $R_f P$, and price informativeness PI , as functions of the bond-market demand shock u^B —keeping the payoff F and the stock-market demand shock u^S fixed (at their expected value). Price informativeness is measured—state-by-state—as the precision of investors' posterior beliefs in excess of prior precision, τ_F , and of the precision of their private signal, τ_ϵ . The graphs are based on the following parameter values: $\rho = 4$, $\mu_F = 1$, $\tau_F = 2.5^2$, $\tau_\epsilon = 1^2$, $\bar{X}^S = 1$, $\tau_{u^S} = 5^2$, $\bar{X}^B = 1$, $\tau_{u^B} = 10^2$, $\alpha = 0$, and $W_{i,1} = 1.5$.

Our discussion so far centred on shocks to noise traders' bond demand. Shocks to the payoff or to noise traders' stock demand do not alter this mechanism but add variability to the price ratio—and hence to price informativeness—unrelated to bond-market learning. Consequently, the relation between the rate of interest and price informativeness is no longer perfectly monotone (see Figure 4) but it does remain strongly positive.

V. Rational Expectations Equilibrium with an Endogenous Interest Rate

We now analyze our main model with initial consumption and an endogenous stock payoff. Most importantly, we demonstrate that the key insights from the illustrative model continue to hold: the bond market reveals discount-rate news, and prices are more informative when the interest rate is high. In addition, we relate the interest rate to allocative efficiency and to asset prices.

V.A. Learning from the Bond Market

The intuition for learning from the bond market again derives from investors' budget constraints and market clearing. In equilibrium, rational investors' stock and bond expenditures plus aggregate consumption (modulo the deterministic component of bond demand, α) sum up to aggregate wealth or, formally:

$$(\bar{X}^S - u^S) P + (\bar{X}^B - u^B) R_f^{-1} + \int_0^1 C_{i,1} di - \alpha = \bar{W}_1. \quad (16)$$

Thus, as in the illustrative framework in Section IV, the aggregate-resource constraint ties together noise traders' stock and bond demands and, hence, the bond market provides a signal, s^B , on noise traders' stock demand, u^S , perturbed by their bond demand, u^B :

$$s^B \equiv -\frac{R_f (\bar{W}_1 + \alpha) - \bar{X}^S R_f P - \bar{X}^B}{R_f P} = u^S + \frac{u^B}{R_f P} - \frac{\bar{C}_1}{P}. \quad (17)$$

The signal now also contains investors' period-1 aggregate consumption, $\bar{C}_1 \equiv \int_0^1 C_{i,1} di$, which—due to their *intertemporal* optimization—is a function of all state variables of the economy.²² Hence, in this more general setting, the signal (17) features not only the noisy stock demand u^S (i.e., discount-rate shocks) but also the fundamental z (i.e., cashflow shocks). Yet, in Appendix C, we show that the information it contains is primarily about the former. As in the illustrative model of Section IV, its informativeness increases in the rate of interest.

Note also that, because of the presence of aggregate consumption, $\bar{C}_1$, the bond-market-clearing condition (17) is also no longer linear in the state variables; accordingly,

²²In particular, aggregate consumption depends on expected trading profits which, in turn, depend on the aggregate squared Sharpe ratio, $\int (\mathbb{E}[F | \mathcal{F}_i] - R_f P)^2 di$ —a nonlinear function of the state variables.

the model must be solved numerically. For that purpose, we extend the numerical solution approach presented in Breugem and Buss (2019) to allow for learning from the interest rate, intertemporal consumption choices, and endogenous output. The algorithm relies on discretizing the state space, which, in turn, allows to explicitly compute investors' posterior beliefs and to exactly solve the first-order and market-clearing conditions. Notably, the algorithm allows for arbitrary price and demand functions; that is, one does not need to parameterize these functions in any form. See the Online Appendix for additional details.

In the following figures, we rely on a specific set of parameter values to illustrate the predictions of our model.²³ We have confirmed that the patterns exhibited in the figures obtain for a wide range of parameter values. Indeed, in *all* our numerical analyses, the impact of bond-market shocks on informational and allocative efficiency as well as on asset prices is as illustrated below. In the Online Appendix, we report some of these additional analyses and, moreover, show that the results also obtain with log and power utility.

V.B. Price Informativeness and the Rate of Interest

As Panel A of Figure 6 illustrates, price informativeness and the interest rate continue to be endogenously correlated. The underlying economic mechanism is the same as in Section IV. That is, keeping fixed the fundamental, z , and the noisy stock demand, u^S , a low realization of the noisy bond demand, u^B , generally leads to a low bond price and, hence, a high interest rate (Panel A of Figure 7).

Investors partly attribute the low total bond demand to a strong demand from rational investors for the *stock* (the cross-market learning effect), and so revise upwards their beliefs about the fundamental (Panel B). These more-optimistic beliefs partially offset the standard discounting effect of a high interest rate, thereby pushing up the price ratio $R_f P$ (Panel C). This, as is apparent from Equation (17), attenuates the error introduced by the bond-market shocks, u^B , in the bond-market signal and, hence, renders the signal more informative about the stock noise. Thus, both the rate of interest (Panel A) and price informativeness (Panel D) are increasing in bond-market shocks.

Figure 7 also illustrates that the association between the interest rate and stock-price informativeness is entirely due to investors' learning from the bond-market signal; indeed,

²³The graphs are based on the following parameter values: $\beta = 0.95$, $\rho = 4$, $\tau_z = 2.5^2$, $\tau_\varepsilon = 1^2$, $\bar{X}^S = 1$, $\tau_{u^S} = 5^2$, $\bar{X}^B = 1$, $\tau_{u^B} = 10^2$, $\alpha = 0$, $W_{i,1} = 3$, $K_1 = 1$, $\kappa = 5$, and $\delta = 0$.

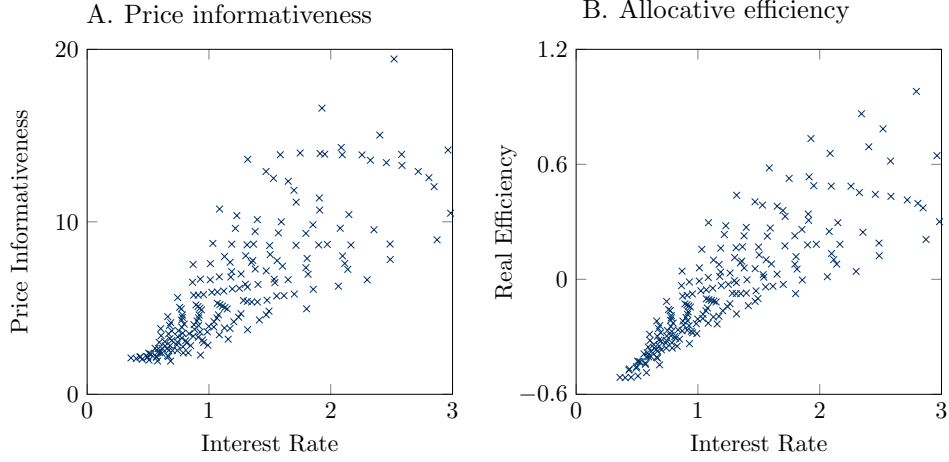


FIGURE 6. INFORMATIONAL AND ALLOCATIVE EFFICIENCY

Notes: The figure plots price informativeness PI (Panel A), and allocative efficiency $\mathcal{E}$ (Panel B), as functions of the rate of interest, R_f . Price informativeness is measured—state-by-state—as the precision of investors’ posterior beliefs in excess of prior precision, τ_F , and of the precision of their private signal, τ_ε . Allocative efficiency is measured—state-by-state—as the “surplus output” that is expected in excess of the output produced by an uninformed manager (cf. Equation (18)). The graphs are based on the following parameter values: $\beta = 0.95$, $\rho = 4$, $\tau_z = 2.5^2$, $\tau_\varepsilon = 1^2$, $\bar{X}^S = 1$, $\tau_{us} = 5^2$, $\bar{X}^B = 1$, $\tau_{u^B} = 10^2$, $\alpha = 0$, $W_{i,1} = 3$, $K_1 = 1$, $\kappa = 5$, and $\delta = 0$.

there is no such association without learning from the bond market (dashed lines). As before, shocks to the fundamental or to noise traders’ stock demand do not alter the mechanism but add variability to the price ratio—and hence to price informativeness—unrelated to bond-market learning. Consequently, the overall relation between the rate of interest and price informativeness is again non-monotone (see Figure 6).

V.C. Implications for Allocative Efficiency, Asset Prices, and Consumption

We turn to the implications of learning from the bond market for allocative efficiency, asset prices, and consumption.

V.C.1 Allocative Efficiency

The firm’s optimal investment, I , is characterized by the standard q -theory investment condition (Tobin 1969): $\frac{I}{K_1} = \frac{\mathbb{E}[z|P, R_f]}{\kappa}$. Specifically, the investment rate, I/K_1 , is driven by the manager’s *conditional* expectation of productivity z , given asset prices. This creates a feedback from financial markets to real investment whereby the stock’s price

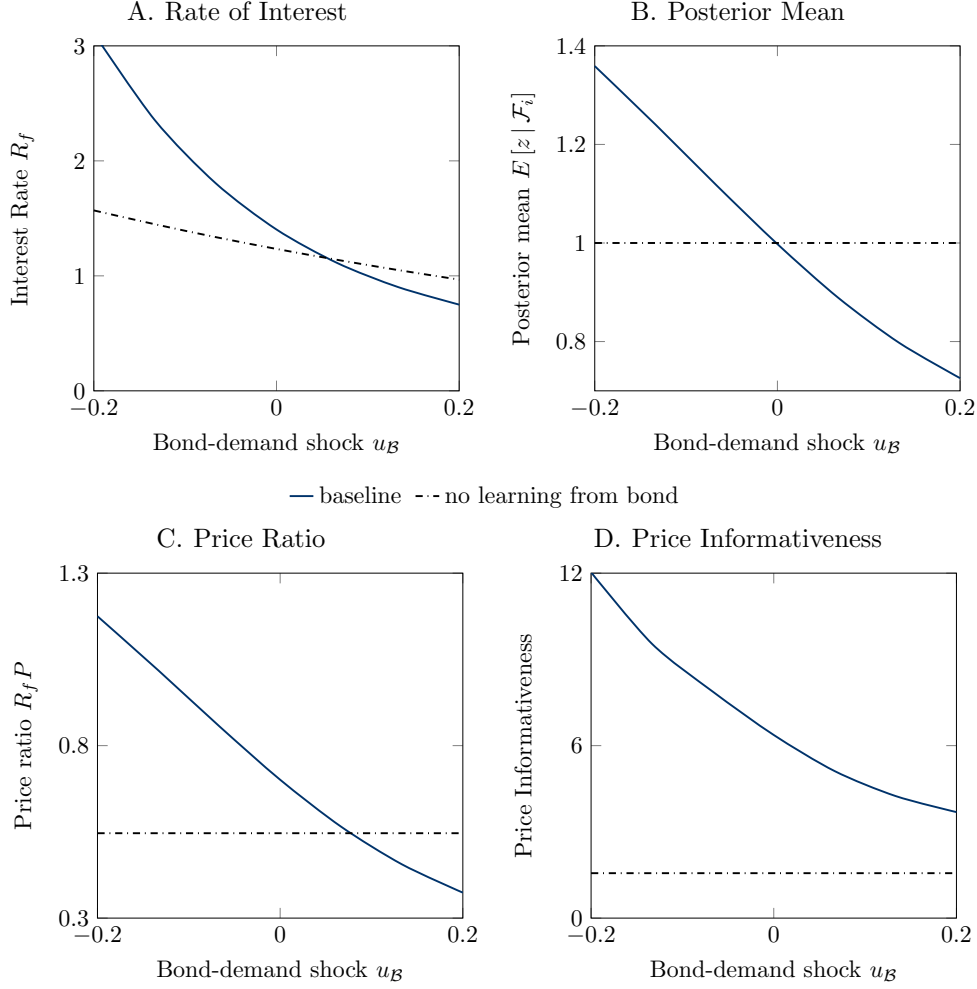


FIGURE 7. IMPACT OF BOND-MARKET DEMAND SHOCKS

Notes: The figure illustrates how bond-market demand shocks, u^B , affect the equilibrium. Specifically, Panels A to D, respectively, plot the rate of interest R_f , investors' posterior mean $\mathbb{E}[z | \mathcal{F}_i]$, the price ratio $R_f P$, and price informativeness PI , as functions of the bond-market demand shock u^B —keeping the fundamental z and the stock-market demand shock u^S fixed (at their expected value). Price informativeness is measured—state-by-state—as the precision of investors' posterior beliefs in excess of prior precision, τ_z , and of the precision of their private signal, τ_ε . The graphs are based on the following parameter values: $\beta = 0.95$, $\rho = 4$, $\tau_z = 2.5^2$, $\tau_\varepsilon = 1^2$, $\bar{X}^S = 1$, $\tau_{u^S} = 5^2$, $\bar{X}^B = 1$, $\tau_{u^B} = 10^2$, $\alpha = 0$, $W_{i,1} = 3$, $K_1 = 1$, $\kappa = 5$, and $\delta = 0$.

(aggregating investors' private information) not only reflects but also affects the firm's value.

A natural measure of allocative efficiency is the “surplus output” that is expected in

excess of the output produced by an uninformed manager (who optimally invests zero):

$$\mathcal{E} = \mathbb{E}[v(z, I) \mid P, R_f] - \mathbb{E}[v(z, 0)]. \quad (18)$$

Panel B of Figure 6 shows that allocative efficiency, $\mathcal{E}$, is positively linked to the rate of interest. Indeed, recall that price informativeness is positively correlated with the interest rate; that is, the higher the rate of interest, the more informative is the stock price about the fundamental z . This, in turn, helps the firm’s manager, who learns from the stock price, to better forecast the productivity shock z and, thus, to invest more efficiently.

V.C.2 Asset Prices and Returns

Learning from the bond market also introduces novel linkages between asset prices, as illustrated in Figure 8.²⁴ First, the stock price displays a inverted U-shape pattern, at first increasing and then declining in the bond-market shock, u^B (Panel A). This is in sharp contrast to the case without learning from the bond market, in which the stock price is monotonically increasing in the bond-market demand shock. Indeed, when expectations regarding the stock’s payoff are held fixed, a lower bond-market demand or, equivalently, a higher rate of interest, imply a stronger discounting and, hence, a lower stock price. With learning from the bond market, this discounting effect is counter-acted by a simultaneous increase in both price informativeness and the posterior mean, which push up the stock price. Consequently, the inverted U-shape pattern arises.

Second, expected returns and return volatility are both related negatively with the rate of interest and positively with bond-market shocks (Panels B and C), as a high price informativeness leads to low expected excess returns (as the risk premium is low) and low excess-return volatility (as the stock’s price tracks its payoff more closely). To the extent that the interest rate is cyclical, both variables are accordingly (endogenously) counter-cyclical. These properties are fully attributable to learning from the bond-market signal, as demonstrated by the comparison with the reference case of an uninformative bond market.

Finally, the bond-stock correlation is stochastic. Indeed, as shown in Panel D, the correlation between stock and bond prices varies endogenously—being positive for high

²⁴For ease of exposition, we concentrate—in this section and the next—on plots displaying bond-market noise shocks, u^B , on the x -axis. Adding fundamental and stock-market noise shocks does not affect any of the results, as demonstrated by the respective scatterplots in the Online Appendix.

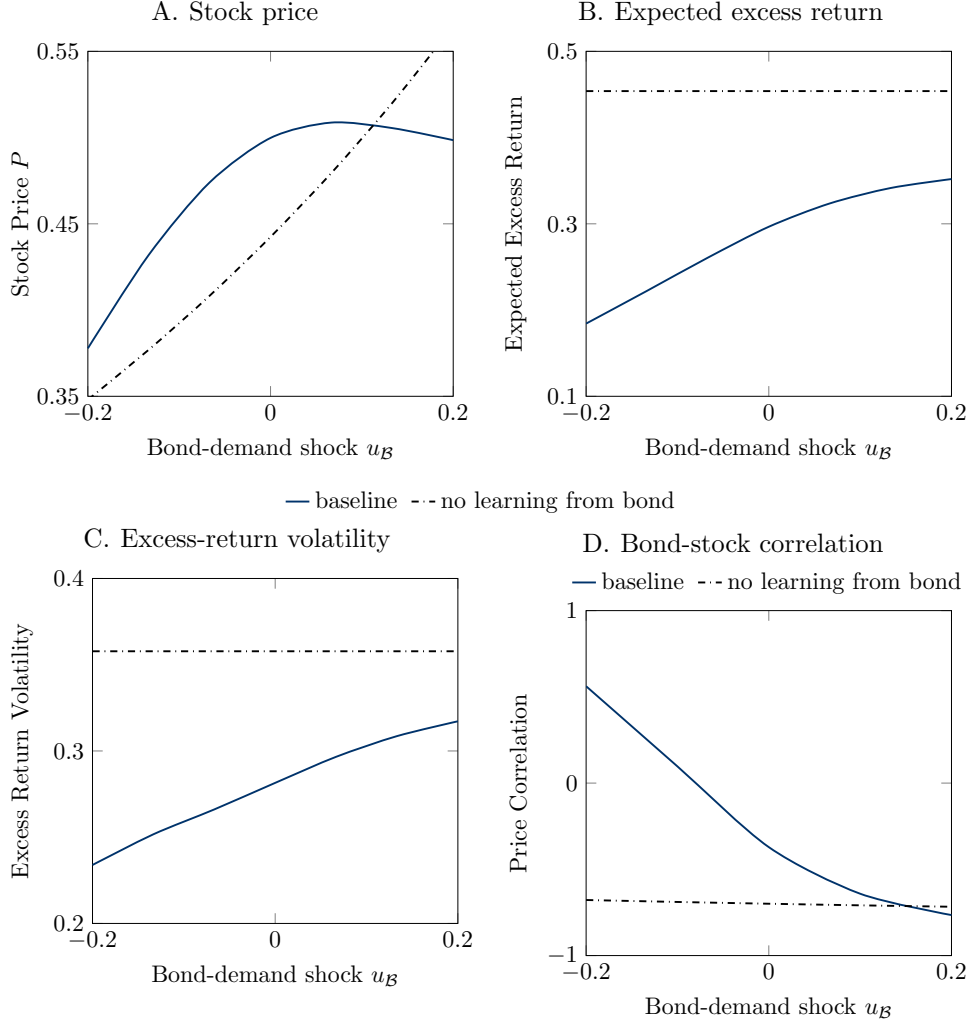


FIGURE 8. STOCK PRICE AND RETURNS

Notes: The figure illustrates how bond-market demand shocks, u^B , affect equilibrium asset prices and returns. Specifically, Panels A to D, respectively, plot the stock price P , the expected excess return $E[F - R_f P]$, excess-return volatility $\text{Vol}(F - R_f P)$, and the bond-stock correlation, as functions of the bond-market demand shock u^B —keeping the fundamental z and the stock-market demand shock u^S fixed (at their expected value). The bond-stock correlation is measured, conditional on u^B , as the unconditional correlation between the stock and bond price. The graphs are based on the following baseline parameter values: $\beta = 0.95$, $\rho = 4$, $\tau_z = 2.5^2$, $\tau_\varepsilon = 1^2$, $\bar{X}^S = 1$, $\tau_{u^S} = 5^2$, $\bar{X}^B = 1$, $\tau_{u^B} = 10^2$, $\alpha = 0$, $W_{i,1} = 3$, $K_1 = 1$, $\kappa = 5$, and $\delta = 0$.

interest rates (low u^B) but negative for low interest rates (high u^B). The figure also shows that this property is largely attributable to bond-market learning.

The result hinges pre-dominantly on how investors interpret price changes caused by demand shocks u^S . Suppose, for instance, that there is a high realization of u^S . The demand pressure results in a high stock price which investors partly attribute to a high

fundamental. Accordingly, they rebalance their holdings from the bond to the stock, which reduces the bond price. This results in a negative correlation between the stock and the bond prices (as the case without learning from the bond market in Panel D illustrates).

When investors learn from the bond market, the precision of investors' beliefs and, hence, the impact of demand shocks u^S , vary with the rate of interest. Indeed, as the information environment improves and, hence, investors make better sense of price changes, the bond-stock correlation becomes less negative. For instance, if the precision is higher (e.g., thanks to a more precise bond signal), then the high stock price is correctly viewed as resulting from demand pressure. Consequently, investors rebalance away from the expensive stock toward the bond, thereby raising its price. Accordingly, both prices are high and, hence, the bond-stock correlation is positive.²⁵

V.C.3 Consumption Choices

Variations in the rate of interest also affect investors' consumption choices, as Figure 9 illustrates. Multiple effects shape those choices. The first are standard consumption-smoothing effects (unrelated to bond-market learning). On the one hand, a higher rate of interest increases the price of period-1 consumption relative to period-2 consumption and, thus, shifts consumption from period 1 to period 2 (substitution effect). On the other hand, a higher interest rate makes investors richer and increases consumption in both periods (income effect). Both effects push up consumption in period 2 but operate in opposite directions for period-1 consumption. For usual levels of (absolute) risk aversion, the income effect dominates, and, hence, consumption increases in period 1. The dashed lines in Panels A and B of Figure 9 illustrate these effects.

Bond-market learning amplifies these effects, as a comparison of the graphs with and without learning from the bond market indicates. Specifically, a higher interest rate improves stock-price informativeness, which, in turn, reduces uncertainty and, hence, tempers investors' precautionary-savings motives. As a result, the bond price drops (interest rate increases) further, which strengthens the consumption-smoothing effects and, hence, raises consumption in both periods. In addition, the improvement in allocative efficiency increases expected output and, thus, consumption in both periods.

²⁵Payoff shocks lead to a positive correlation regardless of the quality of the information environment. Indeed, even under imprecise information, an increase in z raises the stock price, and hence, reduces the bond price through the same portfolio rebalancing channel as does a noise trader shock. Notably, this mechanism strengthens as the precision of information improves, increasing further the correlation at high levels of the interest rate.

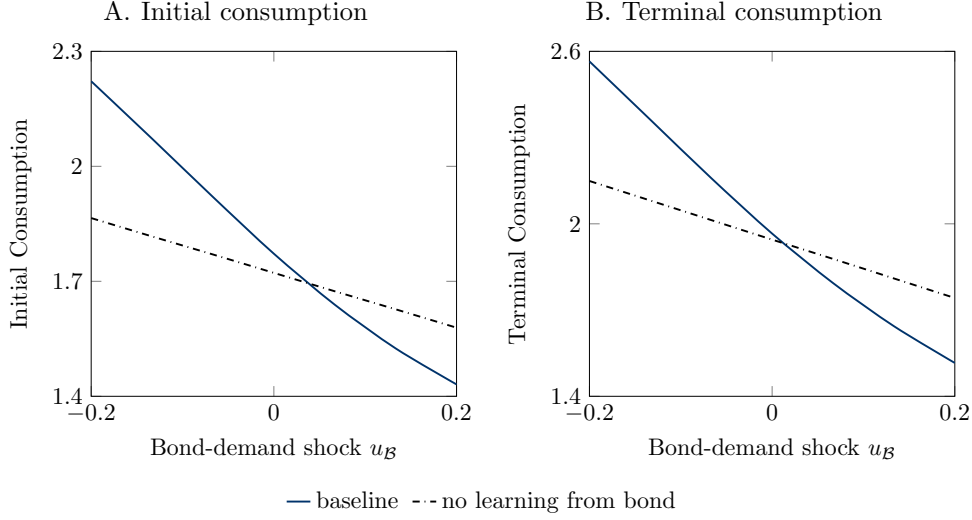


FIGURE 9. CONSUMPTION

Notes: The figure illustrates how bond-market demand shocks, u^B , affect equilibrium consumption choices. Specifically, Panels A and B, respectively, plot initial (period-1) and terminal (period-2) consumption as functions of the bond-market demand shock u^B —keeping the fundamental z and the stock-market demand shock u^S fixed (at their expected value). Consumption is measured for the “median investor,” that is, an investor who receives a signal equal to the realized fundamental. The graphs are based on the following baseline parameter values: $\beta = 0.95$, $\rho = 4$, $\tau_z = 2.5^2$, $\tau_\varepsilon = 1^2$, $\bar{X}^S = 1$, $\tau_{u^S} = 5^2$, $\bar{X}^B = 1$, $\tau_{u^B} = 10^2$, $\alpha = 0$, $W_{i,1} = 3$, $K_1 = 1$, $\kappa = 5$, and $\delta = 0$.

VI. Extensions: Multiple Signals

To illustrate the general applicability of our economic mechanism, we now explore two extensions of our main framework, both of which feature *multiple signals*. As in our main model, investors consume in both periods; the interest rate is determined endogenously, with investors learning from it; and output is endogenous.

VI.A. Multiple Risky Assets

In our first extension, we allow for *multiple risky assets* and focus on the *cross-sectional* implications of learning from the interest rate. The setting is identical to our main model’s, except for the addition of a second stock (for concreteness, stocks can be thought of as industry indices). We denote, respectively, the price of and noise trader demand for stock k by $P^{(k)}$ and u^{S_k} . To highlight the impact of learning from the bond market, we assume that the two stocks are *independent of one another* in terms of fundamentals, private signals, and noise traders’ demand.

To understand the role that learning from the interest rate plays in this economy,

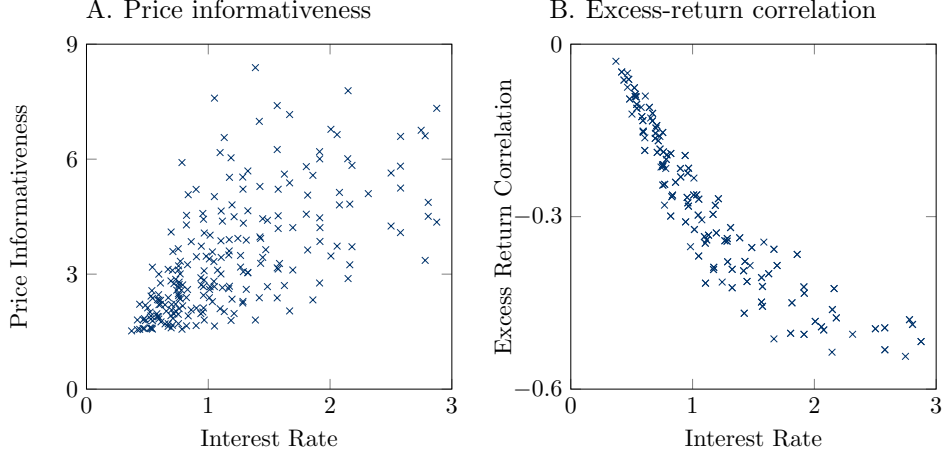


FIGURE 10. EXTENSION: TWO STOCKS

Notes: The figure plots price informativeness (Panel A) and excess-return correlation (Panel B), as functions of the rate of interest, R_f . Price informativeness is measured, for each stock, as the precision of investors' posterior beliefs in excess of prior precision, τ_F , and of the precision of their private signal, τ_ε . Because of symmetry, it is the same for both stocks. The excess-return correlation is calculated as the conditional correlation between the two stocks' returns in excess of the interest rate: $F^{(k)} - R_f P^{(k)}$. The graphs are based on the following parameter values: $\beta = 0.95$, $\rho = 4$, $\tau_z = 2.5^2$, $\tau_\varepsilon = 1^2$, $\bar{X}^S = 1$, $\tau_{u^S} = 5^2$, $\bar{X}^B = 1$, $\tau_{u^B} = 10^2$, $\alpha = 0$, $W_{i,1} = 5$, $K_1^{(k)} = 1/2$, $\kappa = 5$, and $\delta = 0$.

note that aggregating rational investors' budget constraints and clearing markets yields a two-stock version of the aggregate-resource constraint (16). That is, in equilibrium, the aggregate demand for the three assets plus aggregate consumption equal aggregate wealth:

$$(\bar{X}^{S_1} - u^{S_1}) P^{(1)} + (\bar{X}^{S_2} - u^{S_2}) P^{(2)} + (\bar{X}^B - u^B) R_f^{-1} + \bar{C}_1 - \alpha = \bar{W}_1, \quad (19)$$

and, hence, the bond market provides a signal s^B :

$$s^B = \frac{u^{S_1} P^{(1)} + u^{S_2} P^{(2)}}{P^{(1)} + P^{(2)}} + \frac{u^B}{R_f (P^{(1)} + P^{(2)})} + \frac{\bar{C}_1}{P^{(1)} + P^{(2)}}. \quad (20)$$

This has three important implications. First, consistent with the one-stock model, the bond market provides a signal about the noisy stock demands u^{S_k} (and, through $\bar{C}_1$, about fundamentals $z^{(k)}$) which helps investors better forecast fundamentals. As before, the error in this signal originates from the bond demand shocks, u^B .

Second, as Panel A of Figure 10 illustrates, price informativeness and the rate of interest continue to be positively correlated—through the same channel as in the one-

stock model. Specifically, investors attribute a weak bond demand (and, hence, a low interest rate) to a strong demand for the portfolio of the two stocks which, in turn, leads to a higher price ratio, $R_f (P^{(1)} + P^{(2)})$. This attenuates the error created by bond-market shocks u^B in the signal (20) and pushes up price informativeness.

The third, and more novel, implication is that the bond-market signal induces a *negative correlation* between both stocks' excess returns, even though payoffs and noise trading are independent across stocks. This is because the signal s^B provides information about the (price-weighted) *sum* of investors' expenditures in the two stocks, which creates an inter-dependency in investors' beliefs about the two stocks. For instance, consider again the case of a weak bond demand from noise traders' and, hence, a correspondingly high interest rate. Investors infer from the high interest rate that rational investors demand for the sum of the stocks must be strong. This is consistent with a high demand for both stocks, as well as a very high demand for one stock combined with a low demand for the other. But it rules out a low demand for both stocks. Conversely, a low interest rate, caused by a strong bond demand from noise traders, is interpreted as a sign that rational investors' demand for the bond is strong and hence that their demand cannot be strong for both stocks. Accordingly, investors view as less likely the case in which the two stocks' demands have the same sign, which induces a negative correlation between investors' conditional expectations of the two stocks' noisy demands. This, in turn, leads to a negative correlation between investors' conditional expectations of the stocks' payoffs and, hence, excess returns. We highlight again that, in stark contrast to [Admati \(1985\)](#), the correlation of the excess returns of the two stocks is non-zero despite the stocks' payoffs, private signal errors, and noisy demands being independent of each other.

Notably, this effect strengthens with the precision of the bond-market signal. Hence, the correlation between excess returns declines in the rate of interest, as Panel B illustrates.²⁶

VI.B. Multiple Price Signals

In our second extension, we allow for *multiple price signals*. In particular, we demonstrate that other prices, such as the rate of inflation, also reveal discount rate news. For that purpose, we incorporate money into our single-stock framework and, accordingly, distin-

²⁶Because payoffs and demands are cross-sectionally uncorrelated, the correlation of the two stocks' excess returns is zero absent learning from the interest rate. To accommodate a positive correlation between the stocks, as is typically found in the data, one could assume positively correlated payoffs (or liquidity demands).

guish between nominal and real quantities. Specifically, we assume that investors have CARA utility over both consumption and the quantity of the real money balances they hold. As with the other assets, money is assumed to be in finite supply ($\bar{X}^{\mathcal{M}}$) and subject to a random demand shock from noise traders, $u^{\mathcal{M}} \sim \mathcal{N}(0, 1/\tau_{u^{\mathcal{M}}})$ (uncorrelated with other random variables), which prevents the price system from being perfectly revealing. We denote the price of the good in period $t \in \{1, 2\}$ as P_t^G (with P_2^G normalized to 1).

To understand the role that the good's price and the interest rate play in this economy, note that investors' budget constraints, combined with market clearing, imply the following two equilibrium conditions:²⁷

$$s^{\mathcal{B}} = u^{\mathcal{S}} + \frac{u^{\mathcal{B}}}{R_f P} + \frac{u^{\mathcal{M}}}{P_1^G P} - \frac{\bar{C}_1}{P} \quad \text{and} \quad s^{\mathcal{M}} = u^{\mathcal{S}} + \frac{u^{\mathcal{B}}}{R_f P} + \frac{\rho + \omega}{\rho} \frac{u^{\mathcal{M}}}{P_1^G P}, \quad (21)$$

where $s^{\mathcal{B}}$ and $s^{\mathcal{M}}$ represent two distinct signals (and ω is a positive parameter capturing agents' preference for money holdings). The first equation is the counterpart to the bond-market signal (17), to which noise traders' (real) money holdings were added; it is the result of market clearing in the bond market. The second equation—absent from the model without money—is a *money-market signal* that arises from clearing the money market.

Notably, both the bond and money markets provide signals about the noisy stock demand, $u^{\mathcal{S}}$ (i.e., investors' information set has expanded to $\mathcal{F}_i = \{s_i, P, R_f, P_1^G\}$). Hence, the rate of interest but also the good's price or, equivalently, the rate of inflation, P_2^G/P_1^G , reveal discount rate news, which, in turn, allow investors to form more precise beliefs about the fundamental. The signal errors originate from the noisy bond and money demands, $u^{\mathcal{B}}$ and $u^{\mathcal{M}}$. Furthermore, through aggregate consumption $\bar{C}_1$, the bond-market signal also provides information about cashflows. As before, price informativeness and the rate of interest are endogenously positively correlated. Indeed, both price informativeness and the rate of interest are elevated for low $u^{\mathcal{B}}$ realizations, and vice versa for high $u^{\mathcal{B}}$ realizations (see Panel A of Figure 11).

However, there now also exists an endogenous link between the rate of inflation and price informativeness. Specifically, the money-market noise shock, $u^{\mathcal{M}}$, affects not only the price of the good in period 1 and, hence, the rate of inflation, but also the precision of the signals $s^{\mathcal{B}}$ and $s^{\mathcal{M}}$ —through its impact on the stock price P , the interest rate R_f , and the good's price P_1^G . For instance, rational investors partially attribute a high good's

²⁷The equilibrium is defined as previously with the addition of the market-clearing condition for money: $\int X_i^{\mathcal{M}} di + u^{\mathcal{M}} = \bar{X}^{\mathcal{M}}$.

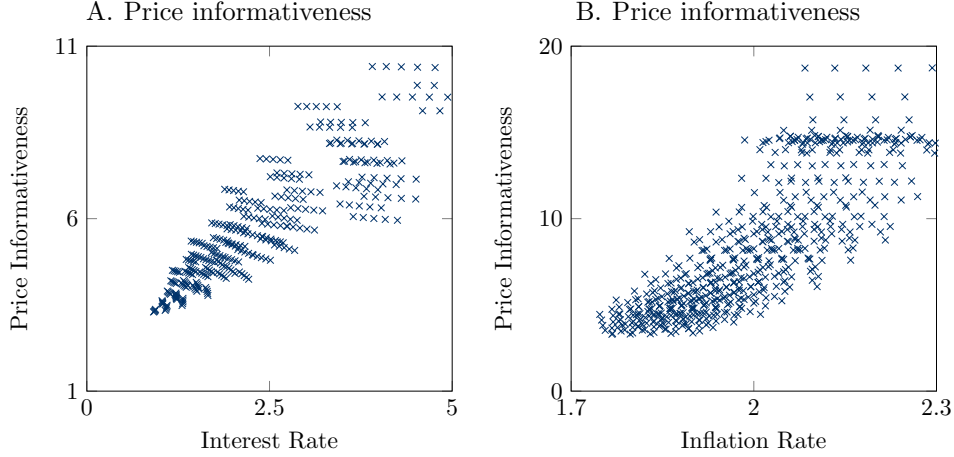


FIGURE 11. EXTENSION: MONEY

Notes: The figure plots price informativeness as a function of the rate of interest R_f (Panel A) and of the rate of inflation P_2^G/P_1^G (Panel B). Price informativeness is measured as the precision of investors' posterior beliefs in excess of prior precision, τ_F , and the precision of their private signal, τ_ε . The graphs are based on the following parameter values: $\beta = 0.95$, $\rho = 4$, $\tau_z = 2.5^2$, $\tau_\varepsilon = 1^2$, $\bar{X}^S = 1$, $\tau_{u^S} = 5^2$, $\bar{X}^B = 1$, $\tau_{u^B} = 10^2$, $\bar{X}^M = 1$, $\tau_{u^M} = 20^2$, $\alpha = 0$, $\lambda = 0.1$, $\omega = 4$, $W_{i,1} = 5$, $K_1 = 1$, $\kappa = 3$, and $\delta = 0$.

price, caused by a weak money demand from noise traders (i.e., by a low realization of the money-market noise u^M) to a weak money demand in aggregate and, hence, to a strong rational demand for the stock (and the bond). From this, in turn, investors infer a high fundamental, which pushes up the price of the stock, P . In our illustration, this effect dominates the simultaneous decline in the time-1 price of the good (i.e., an increase in inflation) and the rate of interest. As a result, the price ratios, $R_f P$ and $P_1^G P$, increase, attenuating the noise in both signals. Accordingly, a negative (positive) link between price informativeness and money-market noise shocks (inflation) arises endogenously. Panel B illustrates this.

Interestingly, price informativeness is linked to *both* the rate of interest and rate of inflation even if learning from one of the two markets is not possible. This is a consequence of the simultaneous variations in both variables caused by shocks in any of the two markets.²⁸ Indeed, any variation affects the precision of the signal in the other (informative) market and, hence, price informativeness. Only if investors cannot learn from both markets, is price informativeness independent of the rates of interest and inflation.

²⁸The positive relation between the interest rate and the period-1 good's price is easily understood when investors derive no utility from real money holdings. In that case, the first-order condition for an investor's money holdings implies that $R_f = P_1^G/P_2^G$.

VII. Application: Policy

In this section, we flesh out the policy implications of our work. Specifically, we discuss how central banks and governments, by acting in the bond market, might affect efficiency in the stock market through our novel cross-market-learning channel.

For that purpose, we extend the framework we laid out in Section III (with initial consumption and endogenous production) by explicitly incorporating a central bank’s demand for bonds. Specifically, we model its bond demand as an exogenous random variable, $y \sim \mathcal{N}(\mu_y, 1/\tau_y)$, that is independent of the other shocks in the economy.²⁹ Variations in the mean and precision of y can be linked to a variety of central bank policies. For instance, quantitative easing (tightening) translates into a high (low, potentially negative) mean demand, μ_y . Similarly, a more transparent (opaque) policy—that is, disclosing a narrower (wider) range for planned bond purchases—translates into a higher (lower) bond-demand precision, τ_y . Note that under these interpretations, the central bank’s policy is exogenous and, accordingly, does not convey any information to investors. Our focus is on how such a policy affects the public’s (as well as the government’s own) ability to learn about economic fundamentals from stock prices.

Figure 12 illustrates how these policies affect price informativeness as well as allocative efficiency. We thereby measure price informativeness, consistent with the empirical proxy we employed in Section II, as the square root of the unconditional variance of the predictable component of the (endogenous) payoff, conditional on prices. Allocative efficiency is measured as the unconditional “surplus output,” that is, the expected output in excess of the output produced by an uninformed manager.

Our main findings can be summarized as follows. First, a higher mean bond demand from the central bank, μ_y , leads to a deterioration in informational and allocative efficiency. Intuitively, the higher resultant aggregate bond demand lowers the rate of interest, which, as previously discussed, hampers investors’ ability to learn from the bond-market signal and firms’ ability to allocate capital. It is important to highlight that the worse allocative efficiency does not result from a lower *level* of investment. Instead, it stems from less *efficient* investment decisions; that is, the manager has a harder time differentiating between high-productivity states (in which she should invest more) and low-productivity states (in which she should invest less).

²⁹It would be straightforward to allow for a correlation between the central bank’s demand and, say, noise traders’. Indeed, emerging countries have engaged in quantitative easing even though their rates are well above zero, in order to offset variations in the (liquidity) demand for long-term bonds; see, e.g., [The Economist \(2020\)](#). Formally, this would simply correspond to a reduction in the variance $\tilde{u}^B$.

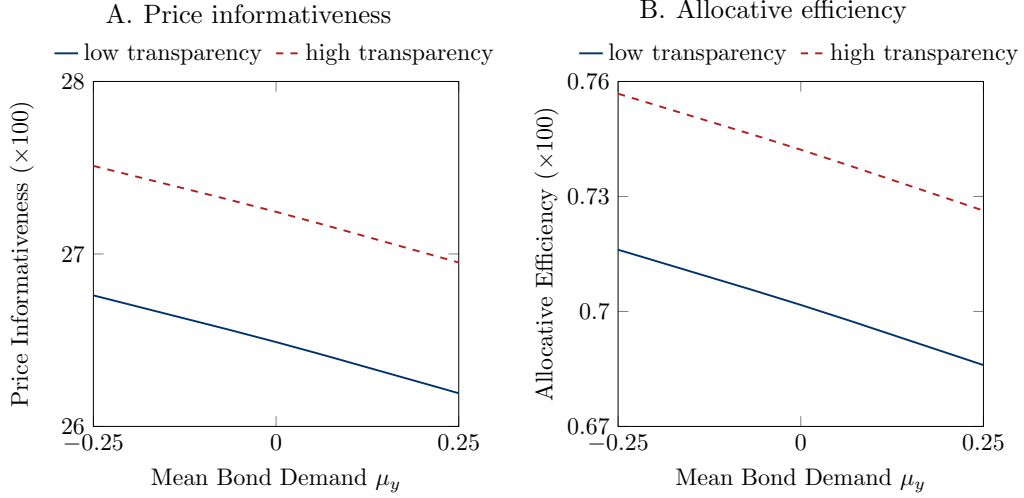


FIGURE 12. POLICY IMPLICATIONS: VARIATIONS IN DEMAND

Notes: The figure plots price informativeness (Panel A) and allocative efficiency (Panel B) as functions of the mean central-bank bond demand μ_y ; for two different levels of demand precision τ_y . Price informativeness is measured as the square root of the unconditional variance of the predictable component of the payoff F , conditional on prices: $\sqrt{\text{Var}(F) - \mathbb{E}[\text{Var}(F | R_f, P)]}$. Allocative efficiency is measured as the unconditional “surplus output” that is expected in excess of the output produced by an uninformed manager: $\mathbb{E}[v(z, I) - v(z, 0)]$. The graphs are based on the following parameter values: $\beta = 0.95$, $\rho = 4$, $\tau_z = 2.5^2$, $\tau_\varepsilon = 1^2$, $\bar{X}^S = 1$, $\tau_{u^S} = 5^2$, $\bar{X}^B = 1$, $\tau_{u^B} = 10^2$, $\alpha = 0$, $W_{i,1} = 3$, $K_1 = 1$, $\kappa = 5$, $\delta = 0$, and $\tau_y = 15^2$ (low transparency) as well as $\tau_y = 20^2$ (high transparency).

Second, an increase in the precision of the central-bank’s bond demand, τ_y , improves informational and allocative efficiency. Intuitively, a higher precision attenuates the noise in the bond market and, hence, renders the bond-market signal more precise.

Finally, our model sheds light on how limits on the rate of interest (be they natural or artificial) affect investors’ ability to extract information from the bond market and, hence, to learn from the stock price. For instance, in the real world, the (gross) interest rate is bounded (from below) by investors’ ability to switch to cash, introducing a natural limit at around one—the “zero lower bound” (which was attained frequently in recent years due to accommodative monetary policy). Similarly, many central banks, such as the Bank of Japan, have engaged in policies to keep interest rates below certain (artificial) limits.

Intuitively, such limits complicate investors’ inference problem and, hence, harm their ability to learn from the bond market. Figure 13 illustrates this point with a (gross) interest rate barred from dropping below unity. Investors who observe an interest rate at the boundary learn only that the unconstrained rate would have been smaller than (or equal to) one but its exact value is censored (that is, many values of the bond-market

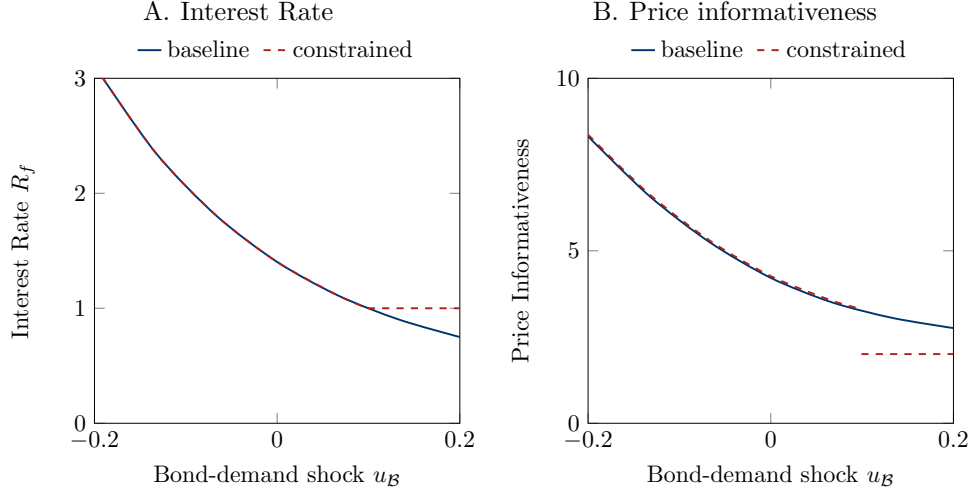


FIGURE 13. POLICY IMPLICATIONS: LIMITS

Notes: The figure plots the interest rate (Panel A) and price informativeness (Panel B) as functions of the bond-market demand shock u^B —keeping the fundamental z and the stock-market demand shock u^S fixed (at their expected value). In the “baseline” case, the (gross) rate of interest is unconstrained, whereas in the “constrained” case it is barred from dropping below unity. The graph is based on the following parameter values: $\beta = 0.95$, $\rho = 4$, $\tau_z = 2.5^2$, $\tau_\varepsilon = 1^2$, $\bar{X}^S = 1$, $\tau_{u^S} = 5^2$, $\bar{X}^B = 1$, $\tau_{u^B} = 10^2$, $\alpha = 0$, $W_{i,1} = 3$, $K_1 = 1$, $\kappa = 5$, and $\delta = 0$.

shock u^B are compatible with the interest rate hitting the boundary). Accordingly, the boundary exhibits a discrete drop in the precision of the bond-market signal and, hence, in price informativeness (as Panel B illustrates).

Overall, these results point towards important consequences of central-bank policies in the bond market for informational and allocative efficiency. Indeed, policies, that affect aggregate demand or restrict interest rate fluctuations might harm stock-market efficiency—through their impact on the investors’ ability to extract information from the bond market. In contrast, more transparent policy implementations can raise informativeness.

VIII. Conclusion

In this paper, we provide new theoretical and empirical insights into how investors use information contained in interest rates to learn about economic fundamentals and how this affects informational and allocative efficiency.

We develop a novel noisy rational expectations equilibrium model in which the interest rate is determined endogenously by supply and demand. We demonstrate that the interest

rate reveals information about noise traders’ stock demand or, equivalently, discount-rate news (plus, potentially, cashflow news). Combined with the stock’s price, this signal allows investors to form more precise beliefs about the stock’s payoff. The strength of the effect is generally positively related to the rate of interest as a higher interest rate attenuates the error in the bond-market signal. Consequently, both stock-price informativeness and allocative efficiency positively correlate with interest rates. The empirical evidence we report lends support to this prediction. The mechanism also endogenously creates counter-cyclicalities in excess returns and in the volatility and comovement of stock returns as well as a stochastic bond-stock correlation—as in the data.

More broadly, our analyses offer novel insights into the impact of fiscal and monetary policies in an environment in which information about economic fundamentals is asymmetric across investors. In particular, we show that increases in the demand for bonds can harm informational and allocative efficiency. As such, our findings highlight novel implications of the strong demand for safe assets observed in recent decades and point towards important unintended consequences of policies such as unconventional monetary policy or “financial repression.” Our findings also highlight that more transparent policies (in the sense of more precise disclosures about, or of stricter commitments to bond purchases) can improve stock-market efficiency.

Finally, our findings also suggest a novel interpretation of the concomitant declines in aggregate productivity growth and real interest rates in the United States (Decker et al. 2017) and in capital allocation efficiency and real interest rates in southern Europe (Gopinath et al. 2017). Specifically, the decline in interest rates might have impaired investor learning about economic fundamentals and, hence, made the allocation of capital less efficient, thereby slowing down productivity growth.³⁰ We look forward to empirical work testing whether these associations are causal or mere correlations.

³⁰Most of this evidence is on private firms, which can be added to our model in a straightforward fashion. Assume that a private firm employs a similar production technology, with productivity that is correlated with the public firm’s (e.g., through economy-wide shocks). The private firm’s manager then learns about the public firm’s productivity from the stock price and the interest rate (just as the public firm manager does), from which she learns about her own productivity. Thus, a higher interest rate facilitates learning for the private firm and increases the efficiency of the private firm’s investments (as for the public firm).

Appendix

A. Microfounding Bond-Market Noise

We think of the price-inelastic bond-market noise as emanating from the demand for home bonds by foreign investors, such as mutual or pension funds, insurance companies, or central banks, who face shocks to their wealth, risk appetite, expectations, or benchmarks.

For instance, consider a foreign investor with CARA-preferences (with absolute risk aversion ρ^* and initial wealth W_1^*) which maximizes expected utility from terminal wealth W^* (denoted in units of its own country's good). Further assume that the period-2 exchange rate, e , is characterized by $e \sim \mathcal{N}(\mu_e, 1/\tau_e)$ and, without loss of generality, that the period-1 exchange is 1. In addition to the home bond (i.e., the bond with interest rate R_f), the institution can invest into a bond from its own country with rate of interest R_f^* . Finally, it is endowed with u^B shares of a non-tradable asset whose payoff is, without loss of generality, perfectly correlated with the period-2 exchange rate e .

Accordingly, the foreign investor's budget constraints (in units of its own good) are:

$$X^B R_f^{-1} + X^{B^*} (R_f^*)^{-1} = W_1^*; \quad (X^B + u^B) e + X^{B^*} = W^*, \quad (\text{A1})$$

where X^B and X^{B^*} denote the investors holdings of home and foreign bonds, respectively.

Maximizing the investor's expected utility subject to the budget constraints (A1), yields the following demand for the home bond: $\frac{\mu_e - R_f^* R_f^{-1}}{\rho^* (1/\tau_e)} + u^B$ and, hence, gives rise to a price-inelastic shock, u^B ; exactly as in our assumption for the bond-market noise.

B. Proofs and Derivations

Section IV: With exogenous output, the fundamental F is normally distributed with mean μ_F and precision τ_F : $F \sim \mathcal{N}(\mu_F, 1/\tau_F)$. Investors receive a private signal $s_i = F + \varepsilon_i$.

We conjecture (and later verify) that the market-clearing conditions in the bond and the stock market (6) are linear in the state variables:

$$0 = b_0 + b_1 u^S + b_2 u^B, \quad \text{and} \quad R_f P = a_0 + a_1 F + a_2 u^S. \quad (\text{A2})$$

Both serve as public signals for investors. In particular, the novel feature of our model is that the bond-market-clearing condition (A2) provides a signal about the noisy stock

demand u^S : $s^B \equiv -\frac{b_0}{b_1} = u^S + \frac{b_2}{b_1} u^B$. This allows investors to form beliefs about u^S *conditional* on signal s^B , with conditional precision $\tau_{u^S|s^B} \equiv \text{Var}(u^S | s^B)^{-1}$ and conditional mean $\mu_{u^S|s^B} = \mathbb{E}[u^S | s^B]$:

$$\tau_{u^S|s^B} = \tau_{u^S} + \frac{b_1^2}{b_2^2} \tau_{u^B}, \quad \text{and} \quad \mu_{u^S|s^B} = \frac{1}{\tau_{u^S|s^B}} \frac{b_1^2}{b_2^2} \tau_{u^B} \left(-\frac{b_0}{b_1} \right). \quad (\text{A3})$$

Combining these conditional beliefs about the noisy stock demand with an investor's prior information about F , her private signal $s_i = F + \varepsilon_i$, and the conjectured stock price signal $R_f P$, in (A2), yields her posterior beliefs about the stock payoff F :

$$\tau \equiv \text{Var}(F | s_i, R_f, P)^{-1} = \tau_F + \tau_\varepsilon + \frac{a_1^2}{a_2^2} \tau_{u^S|s^B}; \quad \text{and} \quad (\text{A4})$$

$$\begin{aligned} \mathbb{E}[F | s_i, R_f, P] &= \frac{\tau_F}{\tau} \mu_F + \frac{\tau_\varepsilon}{\tau} s_i + \frac{\tau_{u^S|s^B}}{\tau} \frac{a_1^2}{a_2^2} \frac{R_f P - a_0 - a_2 \mu_{u^S|s^B}}{a_1} \\ &= \frac{1}{\tau} \left(\tau_F \mu_F - \frac{a_1}{a_2^2} \tau_{u^S|s^B} (a_0 + a_2 \mu_{u^S|s^B}) \right) + \frac{\tau_\varepsilon}{\tau} s_i + \frac{\tau_{u^S|s^B}}{\tau} \frac{a_1}{a_2^2} R_f P. \end{aligned} \quad (\text{A5})$$

In the absence of initial consumption, the objective of investor i is to choose her portfolio holdings in the stock, X_i^S , and in the bond, X_i^B , to maximize the expected utility from terminal consumption $C_{i,2}$ subject to the budget constraints:

$$X_i^S P + X_i^B R_f^{-1} = W_{i,1} \quad \text{and} \quad C_{i,2} = X_i^S F + X_i^B. \quad (\text{A6})$$

Solving the period-1 budget equation (A6) for the bond holdings, X_i^B , yields

$$X_i^B = R_f (W_{i,1} - X_i^S P), \quad (\text{A7})$$

which can be used to rewrite the period-2 budget constraint in (A6) as

$$C_{i,2} = R_f W_{i,1} + X_i^S (F - R_f P). \quad (\text{A8})$$

Plugging the period-2 consumption (A8) into the investor's utility function and maximizing the utility with respect to X_i^S yields the traditional CARA optimal stock demand:

$$X_i^S = \frac{\mathbb{E}[F | s_i, R_f, P] - P R_f}{\rho \text{Var}(F | s_i, R_f, P)}. \quad (\text{A9})$$

Equations (A7) and (A9) jointly characterize investors' optimal portfolios.

Aggregating investors' bond demand (A7) and imposing market clearing in the stock and bond markets implies

$$\begin{aligned} \int_0^1 X_i^B di + \alpha R_f + u^B &= \int_0^1 R_f (W_{i,1} - X_i^S P) di + \alpha R_f + u^B \\ &= R_f \bar{W}_1 - (\bar{X}^S - u^S) P R_f + \alpha R_f + u^B \triangleq \bar{X}^B. \end{aligned}$$

This verifies conjecture (A2) and (by matching coefficients) directly yields

$$b_0 = R_f (\bar{W}_1 + \alpha - \bar{X}^S P) - \bar{X}^B, \quad b_1 = P R_f, \quad \text{and} \quad b_2 = 1. \quad (\text{A10})$$

As a result, investors' conditional mean and precision about the noisy stock demand in (A3) can be written as in Lemma 1.

Plugging investors' posterior beliefs (A4) and (A5) regarding the payoff F into the stock demand (A9), aggregating across investors, and imposing market clearing yields

$$\begin{aligned} &= \int_0^1 \frac{\tau}{\rho} \left\{ \frac{1}{\tau} \left(\tau_F \mu_F - \frac{a_1 \tau_{u^S|s^B}}{a_2^2} (a_0 + a_2 \mu_{u^S|s^B}) \right) + \frac{\tau_\varepsilon}{\tau} s_i + \frac{\tau_{u^S|s^B}}{\tau} \frac{a_1}{a_2^2} R_f P - R_f P \right\} di + u^S \\ &= \frac{1}{\rho} \left(\tau_F \mu_F - \frac{a_1 \tau_{u^S|s^B}}{a_2^2} (a_0 + a_2 \mu_{u^S|s^B}) \right) + \frac{\tau_\varepsilon}{\rho} F + \frac{1}{\rho} \left(\frac{a_1 \tau_{u^S|s^B}}{a_2^2} - \tau \right) R_f P + u^S \triangleq \bar{X}^S, \end{aligned} \quad (\text{A11})$$

which validates conjecture (A2). Finally, matching the coefficients of (A11) to those in the conjecture (A2) and solving the resultant equation system for a_0 , a_1 , and a_2 , yields

$$a_0 = \frac{\tau_F}{\tau} \mu_F + \frac{\tau_\varepsilon \tau_{u^S|s^B}}{\rho \tau} \mu_{u^S|s^B} - \frac{\rho}{\tau} \bar{X}^S, \quad a_1 = \frac{\tau_\varepsilon (\rho^2 + \tau_\varepsilon \tau_{u^S|s^B})}{\tau \rho^2} = \frac{\tau_\varepsilon + P I}{\tau}, \quad (\text{A12})$$

$$a_2 = -\frac{\tau_\varepsilon (\rho^2 + \tau_\varepsilon \tau_{u^S|s^B})}{\tau \rho^2} \frac{\rho}{\tau_\varepsilon} = \frac{\tau_\varepsilon + P I}{\tau} \frac{\rho}{\tau_\varepsilon}. \quad (\text{A13})$$

Theorem 1 readily follows from (a) plugging coefficients (A10) into the conjecture for the bond-market-clearing condition (A2), (b) plugging coefficients (A12) and (A13) into the conjecture for the stock-market-clearing condition (A2), (c) the optimal bond and stock demands (A7) and (A9).

Section V: Solving the period-1 budget constraint in (5) for the bond holdings, X_i^B , yields $X_i^B = R_f (W_{i,1} - X_i^S P - C_{i,1})$. Aggregating investors' bond demand and impos-

ing market clearing in the bond and stock markets implies

$$\int_0^1 X_i^{\mathcal{B}} di + \alpha R_f + u^{\mathcal{B}} = R_f \bar{W}_1 - R_f P (\bar{X}^{\mathcal{S}} - u^{\mathcal{S}}) - R_f \bar{C}_1 + \alpha R_f + u^{\mathcal{B}} \triangleq \bar{X}^{\mathcal{B}},$$

where $\bar{C}_1 = \int_0^1 C_{i,1} di$ denotes aggregate initial consumption. This is equivalent to Equation (16) and can be re-written to obtain Equation (17).

Taking the first-order condition of the expected firm value, $\mathbb{E}[v(z, I) | R_f, P]$, with respect to real investment, I , yields $-1 + \mathbb{E}\left[(1+z) - \frac{\kappa}{K_1} I \mid P, R_f\right] = 0$, which is equivalent to optimal real investment condition.

Section VI.A: The objective of each investor i is to maximize her expected CARA utility (4), conditional on her information set $\mathcal{F}_i = \{s_i^{(1)}, s_i^{(2)}, R_f, P^{(1)}, P^{(2)}\}$ and subject to the budget constraints:

$$C_{i,1} + X_i^{S_1} P^{(1)} + X_i^{S_2} P^{(2)} + X_i^{\mathcal{B}} R_f^{-1} = W_{i,1}, \text{ and } C_{i,2} = X_i^{S_1} F^{(1)} + X_i^{S_2} F^{(2)} + X_i^{\mathcal{B}},$$

where $X_i^{S_k}$ denotes the number of shares of stock k held by investor i .

Thus, investor i 's bond demand is $X_i^{\mathcal{B}} = R_f (W_{i,1} - X_i^{S_1} P^{(1)} - X_i^{S_2} P^{(2)} - C_{i,1})$. Aggregating investors' demand and imposing market clearing in all three markets yields

$$R_f (\bar{W}_1 + \alpha) - R_f (\bar{X}^{S_1} - u^{S_1}) P^{(1)} - R_f (\bar{X}^{S_2} - u^{S_2}) P^{(2)} - R_f \bar{C}_1 + \alpha R_f + u^{\mathcal{B}} = \bar{X}^{\mathcal{B}},$$

which is equivalent to Equation (19) and can be easily re-written to obtain Equation (20).

Section VI.B: The objective of each investor i is to maximize her expected utility

$$-\frac{1}{\rho} \exp(-\rho C_{i,1}) + \beta \mathbb{E}\left[-\frac{1}{\rho} \exp(-\rho C_{i,2}) \mid \mathcal{F}_i\right] - \frac{\lambda}{\omega} \exp(-\omega (X_i^{\mathcal{M}}/P_1^G)), \quad (\text{A14})$$

subject to the budget constraints:

$$C_{i,1} + X_i^S P + X_i^{\mathcal{B}} R_f^{-1} + \frac{X_i^{\mathcal{M}}}{P_1^G} = W_{i,1}, \text{ and } C_{i,2} = X_i^S F + X_i^{\mathcal{B}} + \frac{X_i^{\mathcal{M}}}{P_2^G}. \quad (\text{A15})$$

Combining the budget constraints in (A15), plugging the resultant period-2 consumption into the investor's utility function (A14), and deriving the first-order conditions with

respect to period-1 consumption, $C_{i,1}$, and money holdings, X_i^M , yields

$$\begin{aligned} \exp(-\rho C_{i,1}) &= \beta R_f \mathbb{E} [\exp(-\rho C_{i,2}) \mid \mathcal{F}_i], \text{ and} \\ -\omega \frac{X_i^M}{P_1^G} &= -\rho C_{i,1} + \ln \left(1 - \frac{P_1^G}{R_f P_2^G} \right) - \ln \lambda. \end{aligned} \quad (\text{A16})$$

Substituting out period-1 consumption from first-order condition (A16) (thanks to (A15)), aggregating across investors, clearing the stock, bond, and goods markets, and simplifying yields

$$\left(1 + \frac{\omega}{\rho} \right) \frac{\bar{X}^M - u^M}{P_1^G} = -\frac{1}{\rho} \ln \left(\frac{1}{\lambda} \left(1 - \frac{P_1^G}{R_f P_2^G} \right) \right) + \bar{W}_1 + \alpha - (\bar{X}^S - u^S) P - \frac{\bar{X}^B - u^B}{R_f},$$

which, after defining $s^M \equiv \frac{\rho + \omega}{\rho} \frac{\bar{X}^M}{P_1^G P} + \bar{X}^S + \frac{\bar{X}^B}{R_f P} + \frac{1}{\rho} \frac{1}{P} \ln \left(\frac{1}{\lambda} \left(1 - \frac{P_1^G}{R_f P_2^G} \right) \right) - \frac{\bar{W}_1 + \alpha}{P}$ yields the signal s^M in (21).

In addition, aggregating investors' period-1 budget constraint in (A15) and imposing market clearing in all three markets yields

$$\bar{C}_1 + (\bar{X}^S - u^S) P + (\bar{X}^B - u_1^B) R_f^{-1} + (\bar{X}^M - u^M) (P_1^G)^{-1} - \alpha = \bar{W}_1,$$

which, after defining $s^B \equiv \bar{X}^S + \frac{\bar{X}^B}{R_f P} + \frac{\bar{X}^M}{P_1^G P} - \frac{1}{P} (\bar{W}_1 + \alpha)$, yields the signal s^B in (21).

C. Bond-Market Signal: Impact of Aggregate Consumption

The key difference between the illustrative model in Section IV and the model with early consumption in Section V is the presence of aggregate consumption, $\bar{C}_1$, in the bond-market signal s^B in (17). In the following, we discuss how this affects the information content of the bond-market signal.

Aggregate consumption, $\bar{C}_1$, (and, hence, the bond-market signal) is a (non-linear) function of *all* state variables in the economy— u^B , u^S , and z . Hence, in contrast to the illustrative model, the signal does not provide information exclusively about the stock-market noise, u^S , but also about the fundamental z . However, we find that the bond-market signal, s^B , is considerably more sensitive to stock-market demand shocks, u^S , than to fundamental shocks, z . Indeed, regressions of the signal on the state variables (and their squares) yields estimates that are higher by an order of magnitude for the stock-

and bond-market noises.³¹

In contrast, a regression with the price ratio, $R_f P$, as dependent variable leads to strongly significant coefficient estimates on the fundamental, while the coefficients on stock noise are reduced by an order of magnitude. In other words, the price ratio (respectively, the bond signal) is the primary source of information about the fundamental (respectively, the stock-market noise).

To study the link between the interest rate and the bond-market signal, we add to each regression the three state variables interacted with the interest rate. Most notably, the sensitivity of the bond-market signal with respect to the stock noise increases in the rate of interest. That is, the signal-to-noise ratio of the signal improves as the rate of interest goes up. Similarly, the sensitivity of the price ratio with respect to the fundamental also increases in the rate of interest.

³¹The details of these regressions are reported in the accompanying Online Appendix.

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Internet Appendix for

What do Interest Rates Reveal About the Stock Market?
A Noisy Rational Expectations Model of Stock and Bond Markets

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IA.1. Summary statistics

TABLE IA1—SUMMARY STATISTICS

Variable	Obs.	Mean	Median	Std. Dev.	Min	Max	ρ_1
Price Informativeness PI_5	50	0.049	0.050	0.011	0.023	0.073	0.475
Price Informativeness PI_3	50	0.040	0.040	0.010	0.021	0.062	0.431
Debt/GDP	50	0.369	0.358	0.123	0.209	0.754	0.854
FED MBS Hold./GDP	50	0.005	0.000	0.016	0.000	0.072	0.781
FED Treasury Hold./GDP	50	0.051	0.050	0.013	0.032	0.107	0.584
S&P500 Volatility	50	0.141	0.139	0.053	0.058	0.296	0.334
Cashflow Volatility	50	0.070	0.069	0.006	0.060	0.095	0.558
Real Interest Rate	50	0.026	0.027	0.027	-0.045	0.085	0.741

Note. The table reports summary statistics for our main variables. *Price Informativeness*, PI_h , refers to the coefficient, $b_{t,h}$, of the cross-sectional regression (1) multiplied by the cross-sectional standard deviation of (scaled) stock prices (for forecasting horizons $h \in \{3, 5\}$). *Debt/GDP* is the ratio of the market value of Treasury debt held by the public to U.S. GDP. *FED MBS Hold./GDP* and *FED Treasury Hold./GDP* are the ratio of the Federal Reserve banks' holdings of MBS and Treasury securities, divided by U.S. GDP. *S&P500 Volatility* and *Cashflow Volatility* are measures of volatility of, respectively, the S&P500 returns and of firms' earnings. *Real Interest Rate* is the nominal rate of long-term U.S. government bonds minus expected inflation (estimated using a random-walk model). ρ_1 denotes the first-order autocorrelation. To align price informativeness with bond market characteristics, we sample stock prices at the end of the U.S. government's fiscal year (either June or September). For each firm, we measure accounting variables at the end of the previous fiscal year—typically December—to ensure that the information is readily available to market participants. We adjust earnings using the gross domestic product (GDP) deflator from the Bureau of Economic Analysis (BEA).

IA.2. Numerical Solution Approach

The main difficulty in identifying the equilibrium in the presence of intertemporal consumption choices is that the market-clearing conditions in the stock and bond markets are nonlinear functions of the state variables, with unknown functional forms. As a result, one cannot explicitly compute the investors' posterior beliefs and, hence, cannot find a closed-form solution for the equilibrium. Accordingly, the model must be solved numerically.

For that purpose, we extend the numerical solution approach presented in [Breugem and Buss \(2019\)](#) to allow for learning from the interest rate, two-period consumption, and endogenous output. The approach allows for arbitrary price and demand functions, that is, one does not need to parameterize (conjecture) these functions in any form. Also, it identifies the equilibrium *exactly*, up to a discretization of the state space (which can be made arbitrarily narrow). The algorithm comprises the following four key steps.

First, we discretize the state space into a grid of N_z , N_{u^S} , and N_{u^B} realizations of the random variables z , u^S , and u^B , respectively.³²

Second, we form, for any given grid point $\Omega = \{z_n, u_m^S, u_o^B\}$, $n \in \{1, \dots, N_z\}$, $m \in \{1, \dots, N_{u^S}\}$, $o \in \{1, \dots, N_{u^B}\}$, the system of equations that characterizes the equilibrium. The system comprises investors' first-order conditions with respect to bond and stock holdings, plus the two market-clearing conditions (6) and the optimal real investment condition. Specifically, to accommodate investors' dispersed signal realizations, we form N_S groups of investors ("signal realization groups") for each grid point Ω , with each group receiving a different signal S_s , $s \in \{1, \dots, N_S\}$. Thus, we arrive at an equation system with $N_S \times 2 + 3$ equations, with unknowns: $R_f(\Omega)$, $P(\Omega)$, $I(\Omega)$, and $\{X_s^S(\Omega), X_s^B(\Omega)\}$, $\forall s \in \{1, \dots, N_S\}$ (i.e., $3 + N_S \times 2$ unknowns in total).

Third, we complement the equation system with a set of equations that characterize investors' rational expectations.³³ Specifically, for each signal-realization group s and each "conjectured" level of productivity $\hat{z}_w$, $w \in \{1, \dots, N_{\hat{z}}\}$, we add equations that, under the beliefs of group s and conditional on prices, describe the aggregate demand for the two assets.^{34,35} This requires solving for the optimal asset demands of all signal realization groups and all conjectured levels of productivity, conditional on prices, and aggregating the resultant demands. This adds $N_S^2 \times N_{\hat{z}} \times 2$ equations for each grid point Ω , though many of those are redundant (i.e., can be removed). Based on the conjectured aggregate demands, $\{\hat{u}_w^S, \hat{u}_w^B\}$, each group s can then compute her posterior probabilities (employed in the first-order conditions) for all conjectured levels of productivity $\{\hat{z}_w\}$, $w \in \{1, \dots, N_{\hat{z}}\}$, using the distribution of the noise traders' bond and stock demands.³⁶

Fourth, for each grid point Ω , we solve the resultant large-scale fixed-point problem using Mathematica. We thereby rely on `FindRoot`, which uses a dampened version of the

³²We truncate the realizations of the bond demand, u^B , such that $\bar{X}^B - u^B \geq 0$. This is needed because, under CARA preferences, the equilibrium might not exist for $\bar{X}^B - u^B < 0$, because of the violation of the Inada conditions.

³³If investors' posterior probabilities were exogenous (e.g., a function of private signals or prior beliefs only), one could directly solve the equation system described in step 2. However, under rational expectations, investors' beliefs depend on the prices of the two assets. This dependence gives rise to a fixed-point problem.

³⁴To distinguish between the actual values of productivity and asset demands at a given grid point, $\{z_n, u_m^S, u_o^B\}$, and conjectured productivity and demands, $\{\hat{z}_n, \hat{u}_m^S, \hat{u}_o^B\}$, we denote the latter with a "hat."

³⁵To allow for conjectured productivity to cover a wide range around the actual productivity z_n , we create a separate grid specific to conjectured productivity, with entries $\{\hat{z}_1, \dots, \hat{z}_{N_{\hat{z}}}\}$.

³⁶Formally, the posterior probability of group s for productivity $\hat{z}_{w'}$, conditional on prices and her private signal S_s , is given by $\mathbb{P}(\hat{z}_{w'} | R_f, P, S_s) = \frac{f_{u^S}(\hat{u}_{w'}^S) f_{u^B}(\hat{u}_{w'}^B) f_z(\hat{z}_{w'} | S_s)}{\sum_{w=1}^{N_{\hat{z}}} f_{u^S}(\hat{u}_w^S) f_{u^B}(\hat{u}_w^B) f_z(\hat{z}_w | S_s)}$, where f_z , f_{u^S} , and f_{u^B} denote the *exact* density functions of productivity z , the noisy stock demand u^S , and the noisy bond demand u^B , respectively.

Newton-Raphson method, together with finite differences to compute the Hessian.

We find that the solution of the system is extremely accurate for $N_z = N_{us} = 5$, $N_{u^B} = 7$, $N_S = 31$, and $N_{\hat{z}} = 31$. Further increasing the number of discretization points leaves the solution virtually unchanged. Based on that grid, solving the system of equations for one grid point takes about 0.8 seconds on an Intel Core i7 workstation. Hence, solving it for all grid points requires less than 10 minutes.³⁷

IA.3. Comparative Statics and Robustness

In this section, we provide a brief comparative statics analysis of the main parameters of the model. In addition, we demonstrate that our insights hold for preferences other than CARA; namely, they hold under constant relative risk aversion (CRRA) preferences. Figure IA1 illustrates the results of these exercises.³⁸ Importantly, the observation that, across all four panels, price informativeness depends on (specifically, declines in) the bond-market demand shock, u^B , confirms the central findings of the paper.

Panel A focuses on the impact of prior precisions. First, an increase in the precision of the bond-market noise ($\tau_{u^B} = 11^2$) increases price informativeness as the bond-market signal (17) becomes more informative which, in turn, allows investors to form more precise beliefs about the fundamental. Second, consistent with the notion that signals are strategic substitutes, a reduction in the precisions of the noisy stock demand, ($\tau_{us} = 4^2$) or of investors' private signals ($\tau_\varepsilon = 0.7^2$), lower price informativeness.

Panel B reports the implications of variations in investors' endowments. An increase in initial wealth ($W_{i,1} = 4$) leads to a increase in the demand for the bond and to a lower interest rate and price ratio. The latter then implies a (slightly) decline in stock price informativeness. Endowing investors with units of the stock ($X_{i,0} = 1$)—while keeping the total supply of the goods unchanged—strengthens the bond-learning channel. Indeed, the value of investors' stock endowments, and so their initial wealth, now also depends on interest rates which amplifies the effects (i.e., the curve steepens).

Panel C displays the results of the remaining comparative statics analyses. A decline in risk aversion ($\rho = 3.5$) shifts up the informativeness of the stock price because investors

³⁷To validate the solution approach, we (a) replicate our closed-form solution for the economy without initial consumption (see Section IV), (b) replicate the Hellwig (1980) solution in an economy without learning from the interest rate and without initial consumption, and (c) confirm that the solution converges to the solution without private information as τ_ε converges to zero.

³⁸For ease of exposition, we again concentrate on plots displaying bond-market noise shocks, u^B , on the x-axis. Adding fundamental and stock-market noise shocks does not affect any of the results.

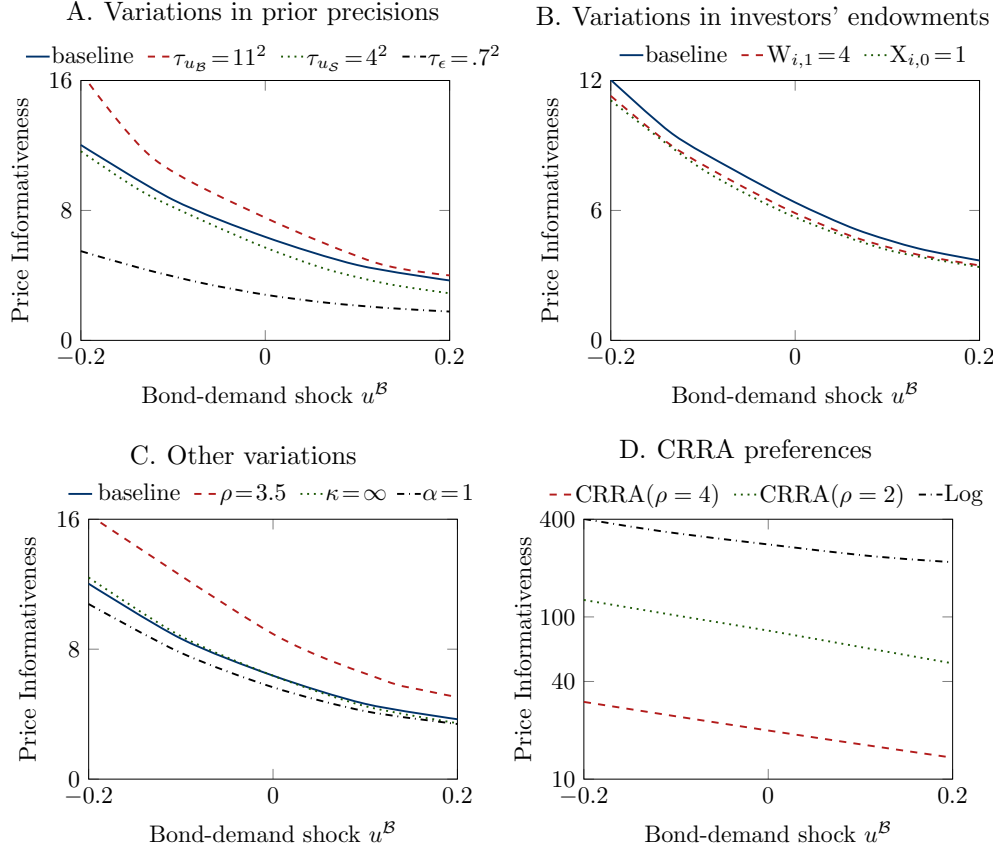


FIGURE IA1. COMPARATIVE STATICS

Notes: The figure plots price informativeness as a function of the bond-market demand shock, u^B —for different variations of our baseline model setup. Price informativeness is measured—state-by-state—as the precision of investors’ posterior beliefs in excess of prior precision, τ_z , and the precision of their private signal, τ_ϵ . The fundamental z and the stock-market demand shock u^S are kept fixed at their expected value. The baseline parameter values are as follows: $\beta = 0.95$, $\rho = 4$, $\tau_z = 2.5^2$, $\tau_\epsilon = 1^2$, $\bar{X}^S = 1$, $\tau_{u^S} = 5^2$, $\bar{X}^B = 1$, $\tau_{u^B} = 10^2$, $\alpha = 0$, $W_{i,1} = 3$, $K_1 = 1$, $\kappa = 5$, and $\delta = 0$, with $\tau_z = 10^2$ for the case of CRRA preferences.

trade more aggressively on their private signals. However, the relative contribution of bond learning weakens (i.e., the curve flattens). This is because it makes the stock price and, hence, the price ratio less sensitive to changes in posterior precisions (thereby weakening the two-way interaction between price informativeness and the price ratio). Making the output exogenous ($\kappa = \infty$) or incorporating a price-sensitive noise component ($\alpha = 1$) leaves the impact of the bond-market signal largely unchanged.

Finally, Panel D demonstrates that our results remain qualitatively unchanged when investors have CRRA preferences. The case of log utility (*Log*) is particularly instructive as it implies a constant wealth-consumption ratio (because the income effect perfectly

offsets the substitution effect). Hence, the signal error originating from aggregate consumption ($\bar{C}_1$) in the bond market signal (17) vanishes, thereby bringing the model closer to the illustrative model discussed in Section IV. Similar to the case of CARA utility, the effect is stronger for low levels of relative risk aversion (note the log-scale for the y -axis).

IA.4. Aggregate-Consumption Regressions

The following table reports the results of regressions of the bond-market signal as well as the price ratio, $R_f P$, on the state variables of the economy. It is referred to in Appendix C.

TABLE IA2—PUBLIC SIGNALS: INFORMATION CONTENT

	s^B (1)	s^B (2)	s^B (3)	$P R_f$ (4)	$P R_f$ (5)	$P R_f$ (6)
z	-1.34 [-3.05]	4.78 [2.55]	1.49 [2.51]	0.71 [60.32]	0.20 [4.63]	0.53 [39.01]
u^S	10.47 [10.79]	10.55 [11.05]	6.15 [5.39]	0.86 [33.18]	0.86 [39.91]	0.81 [31.04]
u^B	14.50 [7.59]	14.69 [7.82]	4.66 [2.10]	-1.98 [-38.91]	-2.00 [-47.23]	-1.79 [-35.21]
z^2		-3.07 [-3.38]			0.26 [12.60]	
$(u^S)^2$		-0.41 [-0.10]			-0.36 [-3.85]	
$(u^B)^2$		-40.45 [-2.43]			1.07 [2.87]	
$z \times R_f$			-0.47 [-3.08]			0.05 [13.43]
$u^S \times R_f$			1.30 [2.54]			0.08 [6.82]
$u^B \times R_f$			1.00 [0.81]			0.15 [5.37]
R^2	0.318	0.347	0.552	0.940	0.960	0.971

Note. The table reports results of regressions of the bond-market signal s^B in (17) and the price ratio, $R_f P$, on the state variables (z , u^S , and u^B), their squared values, and interaction terms with the interest rate, R_f . For each exogenous variable, we consider variations of two standard deviations around its mean. Regressions are estimated using OLS and t -statistics are reported in brackets. The data points are based on an economy with the following parameter values: $\beta = 0.95$, $\rho = 4$, $\tau_z = 2.5^2$, $\tau_\varepsilon = 1^2$, $\bar{X}^S = 1$, $\tau_{u^S} = 5^2$, $\bar{X}^B = 1$, $\tau_{u^B} = 10^2$, $\alpha = 0$, $W_{i,1} = 3$, $K_1 = 1$, $\kappa = 5$, and $\delta = 0$.

IA.5. Additional Figures

Below, we report additional figures that, for brevity, have been omitted from the main part of the text. In particular, Figure IA2 below plots asset prices and returns as well

as investors' consumption choices as functions of the interest rate. Those graphs are reminiscent of Figures 8 and 9 in which the same quantities are plotted against bond-market shocks, u^B . They confirm that shocks z and u^S do not affect any of our take-aways.

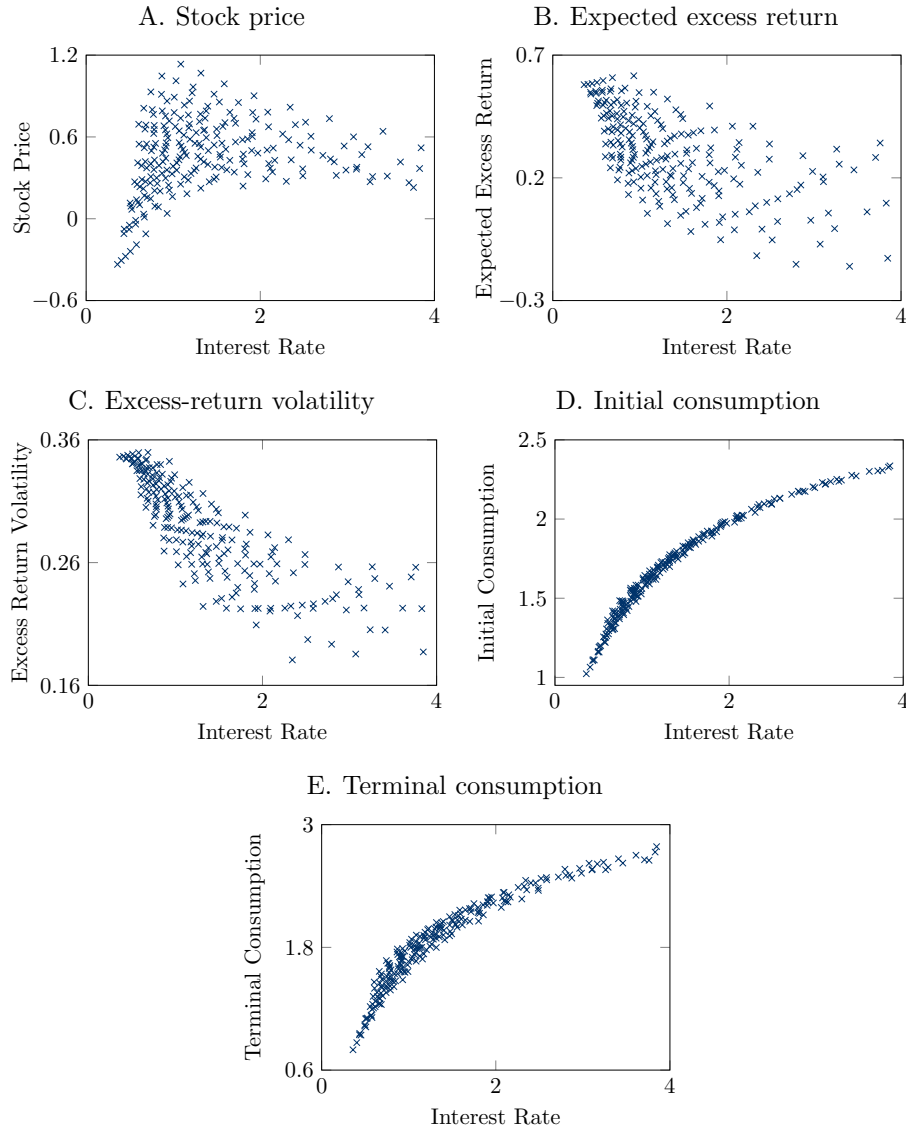


FIGURE IA2. ASSET PRICES AND CONSUMPTION

Notes: The figure plots the stock price P (Panel A), the expected excess return $E[\pi - R_f P]$ (Panel B), excess-return volatility $\text{Vol}(\pi - R_f P)$ (Panel C), initial consumption (Panel D), and terminal consumption (Panel E) as functions of the rate of interest, R_f . Consumption is measured for the “median investor,” that is, an investor who received a signal equal to the realized fundamental.